

CSU – RAMS

RAMSIN Model Namelist Parameters

This document contains details regarding the RAMSIN name-list control file parameters for setting up and running numerical weather simulations using CSU-RAMS v6. This is the primary source of guidance in developing an appropriate name-list pertinent to the weather system being simulated.

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Introduction

This document will describe the input namelist parameters for the RAMS atmospheric model and the RAMS ISAN (Isentropic Analysis) package. It is recommended that the user familiarize themselves with the RAMS Technical Manual before reading this document (even if parts of the Manual are outdated!).

Specifying values for each of the variables in the atmospheric model namelists is the principal way that a user sets up the desired model configuration and selects the many options available for a particular model run. The namelists have the appearance of a standard FORTRAN namelist, and indeed they were patterned after the FORTRAN format. However, there were several deficiencies in the FORTRAN standard, especially in the past, concerning namelists. The biggest deficiency was a complete lack of, or different formats for, comments. Therefore, early in RAMS development, a replacement for the standard FORTRAN namelist reading capability was developed.

The atmospheric model component of RAMS contains several namelists, which have the names:

\$MODEL_GRIDS, \$MODEL_FILE_INFO, \$MODEL_OPTIONS, \$MODEL_SOUND, \$ISAN_CONTROL, \$ISAN_ISENTROPIC.

These namelists are all contained in the data file called **RAMSIN** in the **etc** directory. This is a default name; the input namelist name can be specified as a command line argument. Each namelist in this file begins with one of the above identifiers and ends with the character string "\$END". All variables in these namelists have sample values assigned to them in the **RAMSIN** file, but the user will need to edit many of them for a specific simulation or forecast. In a separate file, **memory/grid_dims.f90**, the user sets values of several different parameters used in dimensioning some of the memory of the atmospheric model. However, the vast majority of the memory is dynamically-allocated. The following sections contain descriptions of all atmospheric model parameters and namelist variables. The descriptions include what the parameters and variables represent or control and provide acceptable or recommended settings for them. Cross-referencing to related variables provides additional information. For this reason, a complete read-through of all namelist variables is recommended.

*Note as a matter of syntax that each **value** assigned to any namelist variable in the **RAMSIN** file must be followed by a comma.*

Model Configuration Parameters – grid_dims.f90

The parameters in the **grid_dims.f90** file are used as array dimensions for many variables in the atmospheric model global memory. These parameters should be reviewed before compilation of the RAMS code. Any time one of these parameters is changed, the entire model code should be recompiled. The parameters set limits on maximum numbers of grid points, grids, etc. Often, the model run is well within those limits. The parameters should be set large enough for the required model space, and for convenience, large enough to fit any anticipated expansions of required model space, but not so large that excessive computer memory is wasted. Most significant memory space is now dynamically allocated, so these parameters are not as important in determining memory usage as in the past. The **grid_dims.f90** parameters are described in the following table.

MAXGRDS integer	The maximum number of grids that may be used in a model run. The number of grids actually used in a run is specified by the namelist variable NGRIDS . NGRIDS may be changed from one model run to the next using the same executable, as long as NGRIDS does not exceed the value of MAXGRDS that was set when compiling the model. Thus, MAXGRDS should be set to the largest number of grids that will be used in a series of runs made from the same compiled code.
NXPMAX integer	The maximum number of grid points in the x-direction to be used on any grid. The actual numbers of grid points spanning the x-direction on individual grids are specified in the multiple values of the namelist variable NNXP . NXPMAX must be equal to or larger than each of the NNXP values.
NYPMAX integer	The maximum number of grid points in the y-direction to be used on any grid. The actual numbers of grid points spanning the y-direction on individual grids are specified in the multiple values of the namelist variable NNYP . NYPMAX must be equal to or larger than each of the NNYP values.
NZPMAX integer	The maximum number of grid points in the z-direction to be used on any grid. The actual numbers of grid points spanning the z-direction on individual grids are specified in the multiple values of the namelist variable NNZP . NZPMAX must be equal to or larger than each of the NNZP values.
NZGMAX integer	The maximum number of vertical levels to be used for the soil model. The actual numbers of levels are specified in the namelist variable NZG . NZGMAX must be equal to or larger than NZG .
MAXSCLR integer	The maximum number of scalars that may be automatically added to a simulation.
MAXDIM integer	MAXDIM must be set to the largest of the values in NXPMAX , NYPMAX , NZPMAX +10, and NZGMAX .
MAXMACH integer	The maximum number of nodes that may be used for a parallel model run. Applies only to running the model in parallel processing mode using MPI.

Model Namelists

The following sections will describe the input namelist parameters for the **RAMS** atmospheric model and the **RAMS/ISAN** (Isentropic Analysis) package.

The following sections contain descriptions of all atmospheric model parameters and namelist variables. The descriptions include what the parameters and variables represent or control and provide acceptable or recommended settings for them. Cross-referencing to related variables provides additional information. For this reason, a complete read-through of all namelist variables is recommended. *Note as a matter of syntax that each value assigned to any namelist variable in the RAMSIN file must be followed by a comma, have no values extending beyond column 80 (although comments can be any length), and no tab characters should be used.* An index of the configuration and namelist variables can be found at the end of this document.

\$MODEL_GRIDS Namelist

The **\$MODEL_GRIDS** namelist provides information to the model primarily on the structure of the one or more nested grids used in a simulation, including location, mesh size, number of mesh points, spatial nesting relationships, time step length, and time and duration of the run. Most of these variables are arrays dimensioned to the parameter **MAXGRDS**. Each value in the array applies to a different grid, several of which may be activated if the user wishes to employ grid nesting. In the **\$MODEL_GRIDS** and other namelists, the multiple values are separated by commas following the variable name and an equals sign. The **RAMSIN** file must have at least as many values of the grid-dependent variables specified as the value assigned to **NGRIDS**, but no more than the value of **MAXGRDS**.

Variable name	Description
EXPNAME character	Character string of up to 128 characters into which the user may put any message (i.e., the experiment name) identifying the particular simulation or run. The character string is written in the standard output listing file and simply serves as a convenient means of labeling the output .
RUNTYPE character	Specifies how the model will run. ERROR, MAKESFC, MAKEVFILE, INITIAL, HISTORY. First, an ERROR run is used for debugging and model development to test initialization and memory allocation. This option will make surface files and output a time=0 analysis file but will stop before running a simulation. In many situations, a RUNTYPE = 'MAKESFC' should be the first run that is made. This type of run will start the model and create surface characteristic files for all grids that will be used during subsequent runs. These characteristics include topography, soil textural class, sea surface temperature, vegetation class, and subgrid distribution of soil textural class, water surface, vegetation type, and NDVI, all of which are available in standard RAMS global datasets from which they can be interpolated onto each model grid. These characteristics will be initialized on the model grids according to the specifications of ITOPTFLG, ISSTFLG, IVEGTFLG, ISOILFLG, and NDVIFLG, then surface files, one for each grid, be created. These surface files will be given a name with a filename prefix specified by the variable SFCFILES and TOPFILES. SST files, one for each grid and for each SST data time available, will be created with a filename prefix specified by the variable SSTFPFX, while a similar set will be created for NDVI using NDVIFILES. These files can then be quickly re-read on subsequent runs of the model or ISAN. An important benefit of the 'MAKESFC' option is that topography for all grids can be defined with a fine grid present, and the fine grid can then be excluded for the first part of the simulation, to be added later on a history restart. This allows the model to utilize topography on its coarser grids that will be consistent with the future addition of the fine grid. If RUNTYPE = 'MAKESFC', the model run will stop immediately after the appropriate files are generated and will not proceed with integration. If RUNTYPE = 'MAKEVFILE', ISAN will perform a data analysis which can include an objective analysis of one or more observational datasets, and produces one or more resultant <i>variable initialization files</i>, or '<i>varfiles</i>'. The varfiles contain a set of atmospheric fields defined on the model grids configured as specified by other variables in the \$MODEL_GRIDS namelist and are to be used in a later run for initializing a simulation, and optionally as 4-D assimilation data during the course of the simulation. The MAKEVFILE option is used only if the model is to be initialized from a complete 3-D objective analysis, (the INITIAL variable in the \$MODEL_GRIDS namelist set to 2) and is the means by which the observational data are processed and interpolated to the model grid(s). The most common way to use this option is to generate varfiles from the objective analysis package every 12 hours, since that is the time interval at which upper-air data are normally collected, or every 6 hours to correspond with the NCEP reanalysis data times. The total number of varfiles required depends on the duration of the simulation to be performed and the particular grids which will be initialized from them (and optionally will use them for 4DDA). The first varfile must correspond to the starting time of the simulation, while the last varfile must equal to or later than the end of the simulation. If RUNTYPE = 'INITIAL', the current model run is designated to be the first of a simulation (see definitions of <i>run</i> and <i>simulation</i> in the entry for TIMMAX below). This means that a simulation is begun at time zero, and that all atmospheric and soil prognostic variables are initialized either horizontally homogeneously from sounding and soil data in the \$MODEL_SOUND and \$MODEL_OPTIONS namelists (INITIAL=1 in this case), read from a varfile prepared in an earlier ISAN run (INITIAL=2), or interpolated from a history file (INITIAL=3). If RUNTYPE = 'HISTORY', the model is to be history restarted, meaning that the atmospheric and soil prognostic

	variables are read from an analysis <i>state</i> file, which was written by the model on a previous run. This option is used when a simulation is carried out over of two or more runs.
TIMEUNIT character	Time units in which the variable TIMMAX is expressed. The allowable values of TIMEUNIT are: 's' to denote 'seconds', 'm' to denote 'minutes', 'h' to denote 'hours', 'd' to denote 'days'. This option allows numerical values for TIMMAX to be specified within convenient ranges for any simulation, which may range in duration from seconds to years.
TIMMAX real	Time during a simulation in units of seconds, minutes, or hours (see TIMEUNIT) when the current run is to stop. We take care here to define the terms <i>simulation</i> and <i>run</i> . A <i>simulation</i> is an entire integration of the atmospheric model from initial conditions to a final time. It consists of one or more <i>runs</i> , which are individual submissions of the model code to the computer for execution. The TIMMAX variable represents the total time elapsed from the beginning of an entire simulation, starting with the first run, and is cumulative over successive runs. However, it denotes the time at which the current run is to terminate, rather than when the entire simulation is to end. If RUNTYPE is set to 'MAKEVFILE' in order to run ISAN, TIMMAX specifies the duration of the time period over which to process observational data and generate varfiles.
IMONTH1 integer	Month of the year when the simulation begins. It is used in conjunction with the namelist parameters IYEAR1 , IDATE1 and ITIME1 to determine the proper solar declination angle for a simulation and to coordinate the model clock with dates and times of various observational datasets and vegetation seasonal cycles. If RUNTYPE is set to 'MAKEVFILE' in order to run ISAN, IMONTH1 specifies the month of the beginning of the time period over which to process observational data and generate varfiles.
IDATE1 integer	Date of the month when the simulation begins. It is used in conjunction with the namelist parameters IYEAR1 , IMONTH1 and ITIME1 to determine the proper solar declination angle for a simulation and to coordinate the model clock with dates and times of various observational datasets and vegetation seasonal cycles. If RUNTYPE is set to 'MAKEVFILE' in order to run ISAN, IDATE1 specifies the date of the beginning of the time period over which to process observational data and generate varfiles.
IYEAR1 integer	Year when the simulation begins. It serves as an identifier of the simulation, along with IMONTH1 , IDATE1 , and ITIME1 to coordinate the model clock with dates and times of various observational datasets. A 4-digit year must be specified for IYEAR1 . If RUNTYPE is set to 'MAKEVFILE' in order to run ISAN, IYEAR1 specifies the year of the beginning of the time period over which to process observational data and generate varfiles.
ITIME1 integer	Coordinated Universal Time (UTC) (or GMT) in hours and minutes (syntax is <i>hhmm</i>) when a model simulation begins (see definition of <i>simulation</i> in the description of TIMMAX). This parameter is used in conjunction with the namelist parameters IYEAR1 , IMONTH1 , and IDATE1 to determine the proper solar declination angle for a simulation and to coordinate the model clock with dates and times of various observational datasets and vegetation seasonal cycles. If RUNTYPE is set to 'MAKEVFILE' in order to run ISAN, IMONTH1 specifies the month of the beginning of the time period over which to process observational data and generate varfiles.
NGRIDS integer	Specifies the number of grids to be activated for the model run. If set to 1, only a single grid covering the entire spatial domain of the simulation will be activated. A setting of 2 will activate a nested grid within the first grid. A nested grid is used to attain a higher spatial resolution in a limited area of the simulation domain. This finer-resolution second grid communicates with the coarser-resolution parent grid via two-way interaction following the scheme designed by Clark and Farley, and Walko et al. Figure 1 shows an example of a lattice consisting of two small, nested grids. The example is a horizontal cross section showing the relative positions of the horizontal velocity components (<i>u</i> , <i>v</i>) and the thermodynamic variables (at locations denoted by <i>t</i>) according to the Arakawa-C grid stagger used in RAMS . (Figure 3 shows a 3D view of the Arakawa-C grid stagger.) Upper-case letters denotes values on the coarse grid, while those on the fine grid are indicated in lower-case letters. NGRIDS can be set to as many nested grids as desired. However, the variable MAXGRDS must be set to a value at least as large as NGRIDS . It is recommended that NGRIDS be set to 1 the first time a simulation is begun, in order to test, at the relatively low cost incurred by the single grid, the remainder of the model configuration specified by the other namelist variables and other data inputs for the particular application. Once these tests are completed, standard procedure is to add finer grids, when necessary, in regions where higher resolution is desired, such as over complex terrain, or when trying to resolve a particular meteorological feature such as a sea-breeze circulation or a thunderstorm. It is common for a given simulation to consist of a series of sequential runs, each new run advancing the simulation forward in time beyond the previous run. RAMS allows the user to increase the value of NGRIDS from one run to the next (on a history restart), thus spawning new fine grids during the course of the simulation, or to decrease NGRIDS to remove one or more grids for the next run. It must be remembered that the variable MAXGRDS needs to be at least the maximum value of NGRIDS which will be used at any time during the simulation.
NNXP integer array	Number of grid cells to span the computational domain in the x-direction. It is an array dimensioned to MAXGRDS , and has a value for each grid activated. These values are written in a single line in the namelist and are separated by commas. The first value in the line refers to grid number 1, the second to grid 2, and so on. The number of grid cells spanning each spatial direction (<i>x,y,z</i>) consists of interior cells where normal evaluation of all terms (such as advective tendencies) in the governing equations is performed, plus a boundary point at each end where variables are computed only by boundary conditions based on interior values. In the example of Figure 1, NNXP (which is NNXP for grid 1) is set to 8, and the first and last grid cells in the x-direction (index values <i>l</i> = 1 and <i>l</i> = 8) are the boundary points. Thus, the actual computational domain may be considered to

	comprise only the interior cells, those shown bounded by lattice lines in the figure, and the user may wish to set NNXP to a value 2 larger than the intended domain size. For the U velocity points, boundary conditions are applied at $I = 1$ and $I = 7$, with normal evaluation of the governing equations being performed at points in between. The U values at $I = 8$ are unused dummy points. The fine grid in Figure 1 also has n_{xp} (which is NNXP for grid 2) equal to 8. (Also see NSTRATX and NINEST). (Figure 3 shows a 3D Arakawa-C grid stagger.)
NNYP integer array	Similar to NNXP but applies to the y-direction. The example in Figure 1 has n_{yp} (NNYP for the coarse grid) equal to 9 and nyp (NNYP for the fine grid) equal to 6. If NNYP is set to 1, the simulation is two-dimensional, and NNYP must then also be set to 1 for all grids activated. A two-dimensional simulation may not be done with NNXP equal to 1 and NNYP greater than 1, <i>i.e.</i> , only x-z cross sections can be used for 2-D runs. However, the x-direction need not be aligned with geographic east but may represent any compass direction with minor changes to the code or unusual placement of the polar stereographic pole point. Pseudo-one-dimensional simulations may be configured by setting NNYP to 1 and NNXP to 5 coupled with cyclic boundary conditions in the x-direction (see IBND). For the example in Figure 1, boundary conditions are applied to T and U at $J = 1$ and $J = 9$, and to V at $J = 1$ and $J = 8$. V at $J = 9$ is a dummy value and is never used. (Figure 3 shows a 3D view of the Arakawa-C grid stagger.)
NNZP integer array	Similar to NNXP but applies to the z-direction. Figure 2 shows an x-z cross section through the model grid analogous to the x-y cross section of Figure 1. Vertical indices for the coarse and fine grids are K and k, respectively, and NZP and n_{zp} (which are the NNZP values for grids 1 and 2, respectively) are the corresponding upper limits of these indices. Vertical velocity components are W and w. The example consists of 7 interior grid cell levels on the coarse grid ($K = 2$ to $K = 8$ of T cells) with boundary cells $K = 1$ below and $K = 9$ above. The ground surface is located at the $K = 1$ W level, while the model top is considered to be at the $K = 8$ W level. It is at this top level where W is set to 0 for the rigid lid model top boundary condition. W at $K = 9$ is a dummy point and is never used. (Figure 3 shows a 3D view of the Arakawa-C grid stagger.)
NZG integer	Number of soil layers to be used in the land surface submodel of RAMS that prognoses energy and moisture in soil, snowcover, vegetation, and canopy air. The soil model consists of a grid representing finite volumes in the uppermost meter or two of the ground. Soil temperature and moisture are prognosed on this grid based on equations governing their internal transport and external exchange with the atmosphere. The horizontal dimensions and resolution of the soil model are identical with those of the atmospheric model grid above, with NNXP and NNYP applying equally to both models. NZG can be set to around 10 for most applications.
NZS integer	Maximum number of snowcover layers allowed to occur in a simulation. It is relevant only when the land surface submodel is activated. NZS must be at least 1 and has no upper limit other than a practical one. If no snowcover is expected in a simulation, there is no need for NZS to be greater than 1. If snowcover will occur, and if its evolution is of some importance to the simulation, a value around 5 is suggested for NZS , to allow multiple snow layers when snowcover is sufficiently deep.
NXTNEST integer array	Identifies the parent grid of each nested grid, <i>i.e.</i> , the coarser grid with which the nested grid has direct communication. Defining values in this array is necessary because different options are available for specifying the interrelation between grids. The rule must always be followed that all grids from 1 up to NGRIDS are numbered consecutively with no integers being skipped, and the coarser grid in which a finer grid is nested always has a lower grid number than that finer grid. We consider first the most common case where the RAMS domain is of limited area, covering no more than about a hemisphere of the earth. For this case, the above rule implies that grid 1 is always the coarsest grid, covering the entire computational domain, and grid 2 is always nested directly within grid 1. Thus, NXTNEST for grid 2 (the second value for NXTNEST) must always be set to 1. However, grid 3 can either be nested in grid 1 (by setting NXTNEST for grid 3 equal to 1), meaning that it is at an equal nesting level alongside grid 2, or grid 3 can be nested within grid 2 (by setting NXTNEST for grid 3 equal to 2), such that it is two nesting levels finer than grid 1. NXTNEST must be set to zero for any grid having no parent: Grid 1 never has a parent, and for a limited area domain, it is the only grid without a parent.
IPRNTSTMT integer	Choice of 0 = No printing, 1 = Yes printing: of extraneous print statements to the standard output. These could be useful for debugging but generally adds clutter to the standard output.
IHTRAN integer	Type of horizontal grid transformation to be used in a model simulation. IHTRAN =0: model grid will be Cartesian, with uniform horizontal spacing used throughout the domain. This option is often used in small computational domains, or where the simulation is of an idealized situation, not corresponding to any particular geographic location. The basic assumption of this grid is that earth curvature effects are negligible. IHTRAN =1: sets up a polar stereographic coordinate system for the simulation. While the model grid spacing itself appears uniform and Cartesian, map factors are activated which cause the earth distances mapped from model distances to be of variable resolution. A polar stereographic projection is a mapping between the spherical earth and a plane (assumed to correspond to zero elevation in the Cartesian model grid) tangent to the earth, with all projection lines emanating from the point on the earth's surface opposite or antipodal to the point of tangency. The Cartesian model plane is centered around the point of tangency, which may be placed anywhere on the earth (the pole in this case refers to the point of tangency, specified in namelist variables POLELAT and POLELON , not to either geographic pole). Thus, size ratio in the center of the model domain is unity between the model grid and the earth, and slowly increases outward toward the model lateral boundaries. This transformation allows a single model grid to cover geographic domains up to hemispheric in size, and avoids the numerical problems encountered near the geographic poles when latitude-longitude coordinates are used.

DELTAX real	Horizontal grid spacing or grid cell size of the model coarse grid (grid 1) in units of meters. This very important parameter determines the scale of meteorological features that are resolvable on the grid, and together with the number of grid cells in the x-direction, NNXP , and y-direction NNYP , determines the total span of the model domain. All spacings on the finer nested grids are derived from this value of DELTAX along with the individual grid spacing ratios NSTRATX . In choosing DELTAX for a simulation, the user needs to carefully consider, among other things, what essential meteorological phenomenon is to be modeled, what the required resolution is for adequately simulating this phenomenon, how large a domain is required to contain the environment of the phenomenon, what the duration of the simulation will be, whether nested grids will be used, and how much computer memory and CPU time are available.
DELTAZ real	Similar to DELTAX , but specifies the vertical, rather than x-coordinate, dimension of a grid cell on the model coarse grid. To be more precise, DELTAZ is the vertical grid spacing of the lowest level of the coarsest model grid (grid 1) and is equal to the vertical grid spacing at higher levels only if constant grid spacing is used in the vertical. (see DZRAT and DZMAX .) However, if DELTAZ is set to 0., it serves as a flag to instruct the model to read the ZZ variable from the namelist for obtaining vertical grid levels directly. As for the case of DELTAX , selecting an optimal value for DELTAZ involves a number of considerations. An additional consideration in this case is that a small value for DELTAZ in a domain containing steeply sloping topography requires high horizontal resolution for computational stability. A rule of thumb here is that the terrain height difference between adjacent grid cells should not exceed about 3-5 times DELTAZ .
DZRAT real	Used in conjunction with DELTAZ and specifies the ratio in vertical grid cell dimension between adjacent levels on the coarse grid. DZRAT is used as a convenient means for vertically stretching the grid, which is a very common practice to obtain high vertical resolution near the ground and lower resolution at higher levels. This method establishes a geometrically-stretched grid, wherein the expansion ratio is constant between consecutive levels. Geometric stretching minimizes the maximum expansion ratio between consecutive levels for a given net stretching over a given number of vertical levels. It is desirable to keep the ratio between consecutive levels small, as large ratios destroy the second-order accuracy of the vertical differencing in the model. Values for DZRAT of 1.1 or even 1.2 are considered acceptable. To illustrate with an example, if DELTAZ is set to 100 and DZRAT is set to 1.1, the first (lowest) grid level will be 100 m thick, the next 1.1 times the first or 110 m, the third 1.1 times the second or 121 m, etc.
ZDELAY real	Height to begin vertical grid stretching. Useful for keeping constant grid spacing in the boundary layer and stretching above the boundary layer top.
DZMAX real	Upper bound on vertical grid spacing anywhere in the model coarse grid and is used in conjunction with DZRAT to prevent the automatic geometric grid stretching from causing overly large vertical grid cell sizes in the upper levels of the model. For example, if DZRAT is set to 1.1, the vertical grid spacing will increase by an order of magnitude over 25 levels. This means that if the first level above the surface is 200 m thick, the 26 th level will be over 2 km thick. By specifying DZMAX to a number such as 1000., the geometric expansion will halt when the 1000 m vertical thickness is reached, and all levels above will be made 1000 m thick.
ZZ real array	Heights of coarsest grid levels in meters, beginning at the ground (ZZ = 0) and continuing to the top level in the model. The number of values specified for ZZ must equal NNZP for grid 1, which is specified in the \$MODEL_GRIDS namelist. Specifying ZZ directly instead of using DELTAZ , DZRAT , and DZMAX is an alternative means of setting the vertical grid structure, and of course offers more flexibility because ZZ can be specified in many ways other than constant or geometrically expanded spacing. However, when using direct specification via ZZ , care needs to be taken not to cause two consecutive levels to have grossly different vertical thicknesses, such as a factor of 2, as this degrades the accuracy of the model's vertical differencing schemes. For the model to use the specified ZZ values, DELTAZ must be set to 0.
DTLONG real	Length of the timestep in seconds at which most processes on the coarse grid will be updated. This is the primary timestep in the model to which all others relate. For example, the values set for NNDTRAT , set integer ratios of timesteps between the coarse and fine grids, such that an appropriate fraction of DTLONG is the actual timestep used on a fine grid. One should set DTLONG reasonably close to, but below the limiting value for computational stability of the model. Given the full model equation set, this value is determined by a number of factors, including wind velocity, internal gravity wave speed, external gravity wave speed (related in part to the height of the model domain), vertical and horizontal grid spacing, and maximum terrain slope. It is not easy to provide a comprehensive formula for estimating an optimal value for DTLONG , and the user should generally try a few values when beginning a new simulation to find a suitable value. However, some guidelines can be provided based on experience. For a deep model domain, say 20 km or more, the model timestep can usually be 5 or 6 seconds for a 1 km horizontal spacing, 30 seconds for a 10 km spacing, 90 seconds for a 60 km spacing, and 180 seconds or more for spacings over 150 km. In a shallow domain 3 km deep, we have been able to set DTLONG to 6 seconds with a 125-meter horizontal spacing. When steep terrain is used (the relevant measure of steepness appears to be the maximum change in terrain height between two adjacent grid cells in vertical grid spacing units, with a value over about 5 being considered steep), DTLONG is often required to be smaller.
NACOUST integer	Number of <i>short</i> or <i>acoustic</i> timesteps performed for each <i>long</i> timestep on a grid. The <i>short</i> timesteps are used for evaluating the pressure gradient force and divergence equation terms and applying them to the velocity and pressure fields, <i>i.e.</i> , the terms involving the propagation of sound waves. The <i>long</i> timestep is used to evaluate most other terms, including advection, diffusion, Coriolis force, and microphysical tendencies. This practice of

	<i>time splitting</i> of the prognostic equations allows full explicit computation (on the short timestep) of the terms governing the rapidly moving sound waves, while the terms governing slower processes can be performed on the long timestep. NACOUST has nothing to do with the relation between timesteps of one grid and another. On the coarse grid, the long timestep is defined by DTLONG, defined above. A separate, grid-dependent variable DTLONGN, which is not defined in the namelists but is internal to the code, is the specific timestep length for each grid. It is equal to DTLONG (defined above) on the coarse grid, and is determined for the finer grids from a combination of DTLONG and NNDRAT, defined below in this Section. The length of the short (sound wave) timestep is twice the ratio (DTLONGN / NACOUST). This short timestep must be short enough (by using a sufficiently large value of NACOUST) that sound cannot travel more than one half the horizontal grid spacing during the short timestep. Sound normally travels more than 300 meters per second, but the model will artificially slow it down if on any grid the short timestep is too large for the grid spacing. A message is printed out stating how much the sound speed has been reduced, with a warning if it is too slow. If this occurs, DTLONGN should be reduced, either by reducing DTLONG, by increasing NNDRAT, or both.
NSTRATX integer array	Number of grid cells in the x-y directions in a fine grid which span the dimensions of a single grid cell of the coarser parent grid to that fine grid. In other words, NSTRATX is the factor of increase in resolution between consecutive nesting levels. For example, if the third value of NSTRATX is set to 4 and the third value of NXTNEST is set to 2, grid 2 will be the parent of grid 3, and grid 3 will have 4 times the resolution or one fourth the grid cell size of grid 2. If in the same example, the second value of NSTRATX were also 4, grid 3 would have 16 times the resolution of grid 1. The example shown in Figure 1 has NSTRATX set to 2 for grid 2. An important constraint between the values of NSTRATX and NNXP or NNYP must be observed for fine nested grids. A fine grid must exactly cover an integer number of grid cells in its coarser parent grid with its <i>interior</i> cells (defined in the entry for NNXP and NNYP) and must have the two additional boundary cells. Thus, NNXP and NNYP for any nested grid must be some integer number (of coarser grid cells covered – 2 in the example of Figure 1) times NSTRATX for that nested grid (3 in the example of Figure 1) plus 2. For NSTRATX and for other related nest configuration variables, if the required combinations of these are not met, the model will stop and issue a message suggesting an acceptable value. Values of 3, 4, and 5 have been most commonly used for NSTRATX. A value of 2 is not usually considered a sufficiently large increase in resolution to make nesting worthwhile, while values of 6 and larger are more prone to cause reflections at the nesting boundaries due to the larger disparity of resolvability between the two grids. (Also see NINEST.)
NNDRAT integer array	Number of timesteps which a fine grid is to be executed to advance it in time the same amount as a single timestep of the coarser parent grid. Because the fine grid has a smaller cell size, it must normally be run with a smaller time step for numerical stability, and the time step length will automatically be made inversely proportional to NNDRAT. As an example, if the second value of NNDRAT is set to 4, and the timestep on grid 1 happens to be set to 20 seconds (see DTLONG), grid 2 will run on a 5-second timestep, four of which will be performed for each timestep of grid 1. The longest timestep on which a grid can be run stably is not directly proportional to the cell size but tends to increase more slowly. Thus, a finer grid nested within a coarse grid at say a 3:1 ratio can normally be run at longer than the expected 1/3 of the coarse grid timestep; 1/2 of the coarse grid timestep may be acceptable.
NESTZ integer	Parameter whose absolute value denotes the number of the nested grid, if any, that is to be nested with a higher vertical resolution than its parent (see NSTRATZ). Such vertical nesting may be done for only one of all model nested grids. Any additional grids that are nested within this grid will use the same vertical grid spacing. Because the vertical nesting ratio can be made variable with height between this nested grid and its parent, the vertical grid spacing of the nested grid may jump suddenly from one vertical level to the next. In order to reduce the relative amount of jump (that is, the ratio of vertical grid spacing between consecutive levels), the model can automatically compensate somewhat by making adjustments to the vertical grid spacings in the parent and coarser grids. This option is activated by giving NESTZ a negative value (of the grid to which it applies). If NESTZ is set to 0, no vertical nesting will be done.
NSTRATZ integer array	Related to NSTRATX but applies to the grid resolution ratio in the z-direction. Unlike NSTRATX, however, NSTRATZ is not a grid-dependent parameter, but refers to the vertical nest ratio between a particular nested grid, specified by the user in the namelist variable NESTZ, and its parent. NSTRATZ is nevertheless a multi-valued parameter, to allow the vertical nest ratio to be varied with height. Each value specified signifies the number of fine grid levels contained in a single level of the parent grid, beginning with the K = 1 level of the parent grid and continuing upward (see Figure 2). Only those levels up to the last one having a vertical nest ratio greater than 1 need be entered in the namelist, because all others are assigned a value of 1 by default. Also, the first value specified in the namelist is not actually used, because it is reassigned in the code. Nevertheless, some value must be entered to serve as a placeholder in the namelist so that the subsequent values are in the proper location. This feature has most often been used to enhance vertical resolution close to the ground on a nested grid, in which case a decreasing sequence such as NSTRATZ = 4, 4, 4, 4, 3, 3, 3, 2, 2, 2, may be used. In this example, the lowest 3 levels of the parent grid will have 4 fine grid levels within each one, the next 3 will have 3 fine grid levels within each one, and the next 3 will have 2 fine grid levels within each one. The remaining levels will have a 1-to-1 vertical nesting ratio. The feature may also be used to enhance vertical resolution in a selected elevated region, such as a cloud layer. Values specified for NSTRATZ must not differ by more than 1 between consecutive entries. Use of vertical nesting is not necessarily the best means of increasing vertical resolution in all cases. The alternative method of increasing vertical resolution over a limited vertical extent of the model domain (but over the entire horizontal extent) has been to stretch the vertical grid spacing (see DZSRAT). For this choice, NSTRATZ could be set to 1, although a combination of both methods of stretching is possible.

POLELAT POLELON real	Geographic latitude and longitude in degrees (ranging from -90 to 90 and -180 to 180, respectively) of the pole point, or point of tangency between the model polar stereographic grid and the earth. This location also serves as the origin ($x = 0, y = 0$) on the polar stereographic grid.
NINEST integer array	Location within a coarser parent grid where the western edge of a nested grid is to be located. Here, the western edge of grid 2 (the western edge always being defined by the lattice line passing through the u-velocity points having an index $I = 1$) is located along the coarse mesh line passing through the U-velocity points having an index $I = 3$. This configuration would be established by setting the second value (for grid 2) of NINEST to 3. Note that the value of NINEST for a nested fine grid always refers to the I-index in the coarser parent grid of a row of U-velocity points. An important constraint involving NINEST , NNXP , and NSTRATX is that the edge of a fine nested grid cannot be any closer than two cells of the coarser parent grid from the lateral parent grid boundary. The example in Figure 1 is right at this limit on both the eastern and western boundaries of the fine grid: NINEST is set to 3, which places the western boundary of the fine grid just two coarse cells from the coarse-grid western boundary, and the eastern boundary of the fine grid is along the $I = 5$ index row of coarse grid U-velocity points, which is just 2 coarse cells away from the coarse-grid eastern boundary at $I = 7$.
NJNEST integer array	Similar to NINEST but locates the southern boundary of a fine grid within the coarser parent grid. Here, the J-index of the coarse-grid V-velocity points is what is referred to by the value of NJNEST . In the example of Figure 1, the second value of NJNEST (for grid 2) would be 3. The constraints described for NINEST likewise apply to NJNEST for the y-direction.
NKNEST integer array	Related to NINEST but locates the bottom boundary of a fine grid within the coarser parent grid. Here, the K-index of the coarse-grid W-velocity points is what is referred to by the value of NKNEST . Unlike the case for NINEST and NJNEST , NKNEST can be set to 1, if desired. Doing so would cause the nested grid to begin at the ground, like the coarser parent grid. If the fine nested grid is to begin at some height above the ground, NKNEST can be set to a larger number. NKNEST can never be set to 2, however; nor can the combination of NKNEST , NSTRATZ , and NNZP be such that the top of a fine nested grid be located exactly one cell of the coarser parent grid below the top of that parent grid. Another constraint is that NKNEST may not be set to 1 for a nested grid if its direct parent grid does not itself begin at the ground. The configuration where a fine nest does not extend to both the ground and the model domain top has been relatively little used. In some cases, vertical motion impinging on a top or bottom nesting boundary of differential horizontal resolution causes strong numerical reflections. In addition, two model algorithms that involve vertical transport, radiative transfer and sedimentation of hydrometeors have not been programmed for transfer between different nested grids. For this reason, a nested grid should begin at the ground and extend to the model top except in a few specialized applications.
CENTLAT CENTLON real array	Grid-dependent parameters for the geographic latitude and longitude in degrees (ranging from -90 to 90 and -180 to 180, respectively) of the center of each model grid. They are used to define the position of the grids in earth coordinates for the purpose of determining the Coriolis force, the solar radiation, and the location of the model domain relative to observed meteorological and land surface data. For the coarsest model grid, CENTLAT and CENTLON are the only means of specifying the center location. Nested grid locations may also be specified relative to their parent grids from variables NINEST , NJNEST , NSTRATX , and NSTRATY . Only if NINEST and NJNEST are given zero values will CENTLAT and CENTLON be used to determine the nested grid location. In such cases, because of the precise positioning of grid lattice points required between a nested grid and its parent, CENTLAT and CENTLON will be only approximate, and the precise location of the nested grid will be the closest allowable position to that which they indicate.
NNSTTOP integer array	Grid-dependent flag to indicate whether each grid extends to the coarse grid top. It must be set to 1 to indicate that a grid does reach the domain top and set to 0 to indicate that the top boundary of a grid is below the model domain top. In view of the recommendation given under the entry for NKNEST , NNSTTOP should normally be set to 1. Of course, since grid 1 always extends to the model domain top, the first value of NNSTTOP must always be set to 1. The specified values of NNZP , NKNEST , and NSTRATZ will themselves determine where the top of a fine nested grid is in relation to the model domain top. The user must ensure that this is in agreement with the setting of NNSTTOP . If not, the model will give a fatal error message and stop.
NNSTBOT integer array	Similar to NNSTTOP . Indicates if a nested grid has a lower boundary at the lower model boundary (the ground) or above it. NNSTBOT must be set to 1 if the fine nested grid begins at the ground, and 0 if it does not. Similar to the case for NNSTTOP , the model will check for agreement between NNSTBOT and NKNEST and will stop with an error message if not in agreement.

\$MODEL_FILE_INFO Namelist

The **\$MODEL_FILE_INFO** namelist consists primarily of variables that control data input to and data output from the model. The names of files containing these data are specified here, as well as the times during a model simulation when the data are read or written. The info in this namelist also controls some aspects of how input data is to be used.

Variable name	Description
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INITIAL integer	Specifies how atmospheric fields in the model are to be INITIALIZED for a new simulation. This should not be changed for a history restart. A history restart is not the same as a history initialization. History initialization is use of RUNTYPE='INITIAL' and INITIAL=3 in which old RAMS output files are used as input for a NEW run. A history restart is simply continuing an existing simulation that needs to be run longer and this uses RUNTYPE='HISTORY' and HFILIN. INITIAL = 1: the atmospheric variables are initialized horizontally homogeneously from a single sounding, which the user supplies in namelist \$MODEL_SOUND. This option is typically used when the model domain is of relatively limited size (a few hundred kilometers across or less), and is required if the model is run in 2-D. Lateral boundaries in this case are open and are updated based on interior values or are held constant in time. INITIAL = 2: the sounding in namelist \$MODEL_SOUND is not used, and instead, the model inputs complete 3-D fields of atmospheric data from <i>varfiles</i>, which have been earlier generated by the model from isentropic output from ISAN (see RUNTYPE). These fields are used both as initial fields in the model and as time-dependent fields to which the lateral boundary region of the model is nudged during integration. INITIAL = 3: the initial fields will be defined by interpolation from a previous RAMS history file, as specified in HFILIN. The new run will start at model time 0. The grids in the current run do not have to have the same structure as the grids on the history file, except that the new grid 1 must fit within the history file grid 1. Each history grid (parent and nested grids) is interpolated to each current grid (parent and nested grids) if the given current grid fits within the given history grid. We could override this and simply interpolate current grid 1 to the nested grids if deemed necessary in the future. This should not be used if a nested grid is added or subtracted; use the standard RUNTYPE='HISTORY' instead.
NUD_TYPE integer	Specifies what type of <i>analysis nudging</i> scheme to use in the 4-dimensional data assimilation (4DDA). The term <i>analysis nudging</i> refers to a scheme where all or a portion of the domain is nudged toward a gridded data analysis. This can also be applied to nudging toward a previous simulation of RAMS or other model output. NUD_TYPE = 0: no analysis nudging is performed. Generally, use this setting if INITIAL=1. NUD_TYPE = 1: analysis nudging is performed using data from the varfiles, with the nudging strengths and settings controlled by the analysis nudging parameters below. Note that varfiles can be created from gridded/observational data using the MAKEVFILE run option, or they can be created from a set of history files using the MAKEHFILE option.
VARFPFX character	Prefix that begins the names of the initialization/nudging files called <i>varfiles</i>, (including path names if applicable) to be read into the model, for cases when the INITIAL=2 (and/or NUD_TYPE=1). These files are output from a previous model run with RUNTYPE set to 'MAKEVFILE' or 'MAKEHFILE' (see VARFPFX). If INITIAL=1, this is ignored. If INITIAL=2 we initialize and nudge with VARFPFX. If INITIAL=3 we initialize from history file but could nudge with VARFPFX.
VWAIT1 real	Used only for automated operational model runs in which an attempt is made to acquire the <i>varfiles</i> data for input to RAMS. If this attempt is unsuccessful and a varfile is not available, the model is instructed to wait a time interval given by VWAIT1 (seconds), before attempting again.
VWAITTOT real	Similar to VWAIT1 and likewise used only for automation of operational runs. Specifies the total amount of time to continue waiting for a completed varfile before giving up altogether.
NUDLAT integer TNUDLAT TNUDCENT TNUDTOP ZNUDTOP real	Parameters that control the option of four-dimensional data assimilation (4DDA) by Newtonian relaxation (nudging). This is also termed <i>analysis nudging</i>, since we will be nudging toward a gridded data analysis. This 4DDA option requires that the INITIAL variable is set to 2 or 3, so that the model is initialized from varfiles or a previous RAMS run. The varfiles or history files contain time series of gridded horizontal wind, potential temperature, and total water mixing ratio values, usually analyzed from observations or sometimes large-scale model forecasts. The atmospheric model solution is relaxed toward the analyzed data during time integration. The strength of the nudging is given by $(I-M)/T$, where I is a varfile data value at a particular location, M is the corresponding model value, and T is a user-specified relaxation (e-folding) time scale. RAMS contains a 3-D array of T values for each grid so that the nudging strength can be specified in a customized way by modifying code. However standard distributions of T may be easily defined through the namelist variables TNUDLAT, TNUDTOP, and TNUDCENT, which define relaxation time scales at the lateral boundary, top boundary, and center regions, respectively, of the model domain. The influence of TNUDLAT extends inward from the lateral boundary of the model domain (coarse grid) by a number of grid cells specified by the user in namelist parameter NUDLAT. The influence function (inverse time scale) increases outward parabolically beginning from the parabola vertex located NUDLAT points in from the boundary. The nudging time scale at that vertex and deeper in the interior of the grid is defined by TNUDCENT. Thus, TNUDCENT can be used to specify a lower bound on nudging strength throughout the domain. The influence of TNUDTOP extends downward from the model domain top to a height specified in namelist variable ZNUDTOP. The influence function increases linearly between these two heights, reaching the minimum value defined by TNUDCENT at and below ZNUDTOP. Note that as

	time scales are inversely proportional to nudging strength, TNUDCENT should have a larger value than TNUDLAT and TNUDTOP for nudging to strengthen toward the lateral and top boundaries. Use of TNUDLAT and NUDLAT is a replacement for the specification of the Davies lateral boundary nudging in 2c and earlier versions of RAMS. Use of TNUDTOP and ZNUDTOP functions analogously to the Rayleigh friction top boundary condition (see descriptions for DISTIM and NUPTS) but can be used for observational data fields which are not horizontally homogeneous. This provides a means for damping upward-propagating gravity waves which could otherwise reflect off the top model boundary. TNUDCENT is a part of the data assimilation capability in RAMS and is definable in the namelist in order to conveniently activate domain-wide nudging. Users of the nudging option should experiment with all three nudging time scales to determine the values which work best for a particular application. The only constraint for numerical stability is that none of the time scales can be less than the model timestep, except that <i>setting TNUDLAT, TNUDTOP, and TNUDCENT to zero turns off the nudging option.</i> Experience has shown that values of TNUDLAT should typically be in the range of 900-1800 seconds, TNUDCENT = 3600 corresponds to very strong nudging and usually is better in the 7200-14400 second range. TNUDTOP frequently does not need to be used and should only be activated at levels well into the stratosphere. Furthermore, if TNUDTOP is used, special care should be taken to examine its impact on the model solution, because the impact can be quite strong.
WT_NUDGE_G real array	Relative weights applied to the nudging weights for each grid. The main intent is to have user control over the 4DDA analysis nudging weights on individual grids. The base weights are determined from the timescale TNUDCENT, then multiplied by WT_NUDGE_GRID. A value of 2., for example, will double the nudging strength on that grid. Care must be taken that all adjustments of nudging strength fall with the numerical stability limits.
WT_NUDGE_UV WT_NUDGE_TH WT_NUDGE_PI WT_NUDGE_RT real	Relative weights applied to the nudging weights for specific variables. The main intent is to have user control over the 4DDA analysis nudging weights. The base weights are determined from the timescale TNUDCENT, then multiplied by WT_NUDGE_UV, etc. WT_NUDGE_UV is applied to the u and v components, WT_NUDGE_TH is applied to potential temperature, WT_NUDGE_PI is applied to the Exner function, and WT_NUDGE_RT is applied to total water mixing ratio. The complete computation of the nudging weight for the analysis nudging 4DDA (in the "center" of the domain) is then computed by: $(WT_NUDGE_GRID(ngrid) * WT_NUDGE_UV) / TNUDCENT$
NUD_COND integer	Flag to activate a condensate nudging 4DDA scheme. This scheme follows the same types of procedures as the history nudging scheme (NUD_TYPE=1), but only nudges total water mixing ratio at grid points where condensate exists. This is intended primarily for assimilation purposes in operational forecast cycles.
TCOND_BEG TCOND_END real	Beginning and ending model time (seconds) to start and stop the condensate nudging scheme.
T_NUDGE_RC real	Timescale for the condensate nudging scheme. The complete computation of the condensate nudging weight is then computed by: $WT_NUDGE_RC(ngrid) / T_NUDGE_RC$
WT_NUDGE_C real array	Relative weight of the condensate nudging scheme for each grid.
IF_ODA integer	Flag to active the ODA (observational data assimilation) feature, a generalized <i>observational nudging</i> scheme. The ODA scheme will examine each station, interpolate in time to each timestep (if the observations are close enough), then perform a "kriging" interpolation to produce three-dimensional value and covariance fields. The combination of the value and covariance fields will only nudge the model fields in locations where the observations are "close enough". The parameters below will control the behavior of the ODA scheme.
ODA_UPA_PREF ODA_SFC_PREF character	File name prefixes (including path information) for the input surface (SFC) and upper air (UPA) data for the ODA scheme. The file must be standard RALPH II format and currently must contain one data time per file.
FRQODA real	Frequency (seconds) at which to update the interpolated observational value and covariance fields with the kriging scheme. This could be done every timestep, but probably less often for efficiency. 10-20 minutes should be an adequate compromise with standard synoptic observations, but it is dependent on the specifics of the simulation and the available observations.
TODABEG TODAEND real	Beginning and ending model times (seconds) to start and stop the ODA scheme.
TNUODA	Main nudging ODA timescale (seconds). This determines, along with WT_ODA_GRID and WT_ODA_UV, etc., the strength of the nudging. For example, the complete computation of the weight for the ODA

real	nudging for the u and v components is computed by: $(WT_ODA_GRID(ngrid) * WT_ODA_UV) / TNUDODA$
WT_ODA_GRID real array	Relative weights applied to the ODA nudging weights for each grid. See TNUDODA .
WT_ODA_UV WT_ODA_TH WT_ODA_PI WT_ODA_RT real	Relative weights applied to the ODA nudging weights for specific variables. WT_ODA_UV is applied to the u and v components, WT_ODA_TH is applied to potential temperature, WT_ODA_PI is applied to the Exner function, and WT_ODA_RT is applied to total water mixing ratio. See TNUDODA .
RODA_SFCE RODA_SFC0 real array	Radii (meters) for the kriging scheme to control the smoothness of the analysis at the surface. SFC0 is the distance from an observation where the influence becomes zero, while SFCE is the distance where the influence drops by a factor of e^{-2} . These values are grid-dependent.
RODA_UPAE RODA_UPA0 real array	Radii (meters) for the kriging scheme for the upper air observations to control the smoothness of the analysis in the upper air. UPA0 is the distance from an observation where the influence becomes zero, while UPAE is the distance where the influence drops by a factor of e^{-2} . These values are grid-dependent.
RODA_HGT real array	The height (meters) above the ground used to determine the vertical structure of the kriging radii. RODA_SFC0 will apply at the ground, while RODA_UPA0 will begin at a height of RODA_HGT . The radius will be held constant at a value of RODA_UPA0 from RODA_HGT to the model top, while a linear interpolation from the ground to RODA_HGT will be performed. Typically, some height above the expected boundary layer depth would be chosen. This value is grid-dependent.
RODA_ZFAC real	Vertical exaggeration factor to control smoothness in the kriging scheme. This is somewhat related to the ratio of horizontal grid spacing and vertical grid spacing. A value of the order 100 is reasonable.
ODA_SFC_TIL ODA_SFC_TEL real	Time interpolation limit (TIL) and time extrapolation limit (TEL) (both seconds) for the surface data in the ODA scheme. These are used to determine the use of the observations at the time of the kriging scheme updates. At an update time, each station will be examined. A past and future observation time (with non-missing data) relative to the update time will be found. If the future minus the past time is less than the TIL, then the data values will be interpolated to the update time for use in the data analysis. If this condition is not true, then the past and future times will be checked to see if it is less than the TEL. If, for example, the past time is less than the TEL, then that data value will be used in the analysis. For hourly surface observations, reasonable settings might be ODA_SFC_TIL =7200., and ODA_SFC_TEL =900.
ODA_UPA_TIL ODA_UPA_TEL real	Time interpolation limit (TIL) and time extrapolation limit (TEL) (both seconds) for the upper air data in the ODA scheme. See ODA_SFC_TIL . For standard 12 hourly surface observations, reasonable settings might be ODA_UPA_TIL =43200. And ODA_UPA_TEL =3600.
HFILIN character	Analysis <i>state</i> file to be read for initializing the current run (RUNTYPE='HISTORY'). This file is also used for IPAST recycle feature (i.e., IPAST_SFC) and History Initialization option (RUNTYPE='INITIAL' with INITIAL=3). The character string placed in this variable must be the full name of the <i>state</i> header file, including the appropriate path if applicable. Further details on the standard syntax of the history file name are given in the description of AFILOUT .
IPAST_SFC integer	Flag to activate a "recycle_sfc" feature, where surface fields from a previous analysis file may be read upon initialization of the current run. All soil and vegetation variables are read, and any default initialization values are replaced. Works similar to History Initialization, but only surface variables are recycled. These do not override surface characteristics in surface files. Allowed values are 0=off and 1=on.
ICLOBBER integer	Flag indicating what the model should do if various files already exist from a prior run, and the model is about to write new files of the same name in the same directory. If ICLOBBER = 0, the model will stop and not overwrite the older files. If ICLOBBER = 1, the model will continue and will write new files, overwriting the old ones.
IOUTPUT integer	Flag controlling whether the model will output any analysis files. If IOUTPUT = 0, no files will be generated. Setting IOUTPUT = 1 causes the model to generate output files in HDF5 format, dependent on other parameters below. The semantics for the output files has changed for v6.0. Previously, we had <i>history</i> files and several types of <i>analysis</i> files. With the use of HDF5, history files are no longer necessary. RAMS now will produce several types of <i>analysis</i> files written in HDF5. These types are: <i>State</i> files – files which have all information of the current model state. These are analogous to a combination of the previous history and analysis files (A and H files). Because of HDF5, these files are

	now smaller than the previous analysis files and contain full precision. <i>Mean</i> files – time-averaged files containing the same variables as on the state files. <i>Lite</i> files – files which contain a user-defined subset of the model fields. <i>“Both”</i> files – time-averaged, lite files The output frequency and other parameters controlling these file types are specified in the variables below.
AFILEPREF character	Prefix for the analysis files output from the model. AFILOUT is used as a prefix to the one or more file names, making up the first part of each name, including directory path information. The model itself appends the last part of each analysis file name, according to the file type, the time during the simulation when the file is written, and the grid number. Separate analysis files are written for each grid number, as well as for each write time. In addition, <i>header</i> files are written at each analysis file write time. As the model appends both the time and the grid number onto the history file prefix (given by AFILOUT), each file name written during a simulation has a distinct name, but all still share the common prefix. AFILOUT is only relevant if IOUTPUT = 1, indicating that files are to be created.
FRQSTATE real	Time interval in seconds at which successive <i>state</i> files are to be output from the model. The model variable TIME begins at zero at the start of a new simulation (and at the TIME value contained in HFILIN for a history start) and counts up to the value of TIMMAX for each run. (see RUNTYPE, TIMSTR, and TIMMAX). When TIME reaches a multiple of FRQSTATE, a model <i>state</i> file is written. In addition, a <i>state</i> file is written out when a model run terminates, even if TIMMAX is not set to an exact integer multiple of FRQSTATE. A <i>state</i> file is also written out when a model run begins, even if the TIME variable is not a multiple of FRQSTATE. FRQSTATE can have a different value for each nested grid. There are times when you may want less frequent output for the coarse grids as compared to the nested grids. So, for example, you can specify: FRQSTATE =7200., 3600.,1800., for a three-grid simulation. If you do not specify a value for each grid, FRQSTATE will default to the previous grid's value (not the parent grid, but the previously specified nest). So, if you specify: FRQSTATE =7200., for a three-grid simulation, all three grids will use the 7200. Second frequency. Although not required, it is recommended to have the higher frequency grids output be an exact multiple of the coarser grids, which should be an integer multiple of the timestep length.
FRQST_KEEP real	Frequency at which the <i>state</i> files are kept. The <i>state</i> files are written at the frequency of FRQSTATE, but any files not written at the frequency of FRQSTATE_KEEP will be deleted after the subsequent file write. This option would be used if more frequent checkpointing of model files are desired. For example, you may have FRQSTATE=900., and FRQSTATE_KEEP=3600. RAMS will write the <i>state</i> files every 900 seconds. So, a set of <i>state</i> files will be created at time=0. And time=900. Seconds. At time=1800 seconds, a new set of files will be created, then the files that were written at time=900 seconds will be deleted. It is intended that FRQSTATE_KEEP be an integer multiple of FRQSTATE, as it makes no sense to have it smaller than FRQSTATE. If FRQSTATE_KEEP=0., no files will be deleted.
FRQLITE real	Similar to FRQSTATE but specifies the time interval between successive writes of 'lite' analysis files. The lite files are intended for writing variables at much more frequent intervals than analysis files, for use in post-processing applications such as graphical display, but with much less information. For example, a user may need surface variables to be output frequently, but not three-dimensional upper air fields. Only the required fields would be specified for output in the lite files.
ITRUNCLITE integer	Flag to permit truncation of LITE file variables to a certain decimal precision. This significantly reduces the LITE file size by only saving the precision you need. This allows more HDF5 compression. The exact decimal precision can be viewed and changed in "src/lib/utlis_f.f90" in subroutines "trunc_2d_vars" and "trunc_3d_vars". Flag = 0 (truncation off) or flag = 1 (truncation on).
NLITE_VARS integer	Number of variables to write to the "lite" analysis file. The actual variable names are specified in the variable LITE_VARS.
LITE_VARS char array	Variable names of the fields that will be written to the "lite" analysis files. The variable names correspond to the RAMS' internal character ID strings, which now can be found in the memory/mem_*.f90 files. For example, the u-component is "UP", potential temperature is "THETA", etc. Any variable that is in memory may be included, but care must be taken in parallel runs to ensure that the field has been passed back to the master process.
ACC_LT_VARS char array	Accuracy of the lite vars if using ZFP/lossy compression. Use 0.0 for lossless compression. Using nonzero values requires compiling with -DENABLE_ZFP_COMPRESSION on the C compiler. The values here MUST be listed in the same order as the variables in flag LITE_VARS.
AVGTIM real	Averaging time period for special time-averaged arrays that may be output to analysis file. Positive values for AVGTIM cause the file write time to be centered in time within this averaging period, while negative

	values cause the file write time to occur at the end of the averaging period.
FRQMEAN real	Similar to FRQSTATE but applies to the output frequency of the 'mean' analysis files.
FRQBOTH real	Similar to FRQSTATE but applies to the output frequency of the 'both' analysis files.
TOPFILES SFCFILES SSTFPFX NDVIFPFX character	Filename prefixes (including path information) for the surface files used in a model simulation. These files contain topography, soil textural class, vegetation type, subgrid distribution of soil textural class, vegetation type, water surface areas, sea surface temperature, and NDVI defined on each model grid. Surface files are always used in a model run. These data types are available on standard RAMS data files defined globally on a latitude-longitude grid and with climatological values for each month of the year for SSTs. Data in surface files may be interpolated from these standard files, from other data sources, or may be defined by other means. These choices are controlled by ITOPTFLG , ISSTFLG , IVEGTLG , ISOILFLG , NDVIFLG . They surface files are commonly made in a separate model run that does not proceed with prognosing atmospheric fields by setting RUNTYPE to 'MAKESFC'. However, if RUNTYPE is instead set to 'MAKEVFILE' or 'INITIAL', and if surface files do not exist or are found to be inconsistent with model grid size, location, etc., new ones will be generated. If they already exist from a prior run, surface data is read from them. A separate surface file is generated for each grid, and the grid number is appended to each filename.
ITOPTFLG integer array	Grid-dependent flag specifying how terrain height data is to be obtained or computed for each model grid. ITOPTFLG = 0, (allowed for a nested grid only), topography for that grid will be interpolated from its parent grid. This will result in relatively smooth topography on the nested grid. ITOPTFLG = 1, topography will be interpolated to the grid from a standard RAMS topography dataset, currently available at 30 arc-second intervals of latitude and longitude for all land areas of the globe. A prefix for all filenames in this dataset is specified in ITOPTFN. This is by far the most common way to initialize topography in RAMS unless the experiment requires idealized or flat topography. ITOPTFLG = 2 – the model will call subroutine TOPTINIT in the model file <i>mksfc_top.f90</i> to assign the terrain height, which by default sets it to zero. To override any of these choices, subroutine TOPTINIT_USER in the file <i>mksfc_top.f90</i>, which is always called but contains only commented out executable lines in its standard form, may be modified. The <i>ruser.f90</i> file is specifically intended for user modification when such a need arises. Subroutine TOPTINIT_USER is normally constructed with an IF block checking for the grid number (NGRID). The terrain height values must be assigned for the appropriate grid number. It must be pointed out that if nested grids are used, it is required that the terrain heights be compatible between different grids. The compatibility requirement is two-sided: Fine grid values corresponding to a single coarse grid cell must locally average to the coarse grid terrain height value in that cell, and the fine grid values near the lateral boundary of the fine grid must be those which would result from a bi-quadratic interpolation from coarse grid values, according to the standard nesting procedure. The model is designed to make the terrain heights compatible automatically and will alter the values assigned to a grid in subroutine TOPTINIT or TOPTINIT_USER if they are not already compatible between grids. A special word of caution is in order here. If a fine grid that is to contain fine resolution terrain height data is to be spawned on a history start at a time after the initialization time of a simulation, the terrain height data on the coarser grids must already have been compatible with it in the preceding portion of the simulation. This may be done either by simply initializing the topography of the spawned grid from its parent, or by generating surface files for all grids, including the one to be spawned, before the simulation started. The latter choice provides for the better-resolved topography on the spawned grid but will also result in very fine (at the 2-delta-x scale) features on its parent grid. This may cause some model noise problems during that earlier part of the simulation on the portion of the coarser grids where the fine grid will be later inserted, but this is normally at acceptable levels (see TOPTWVL).
ISSTFLG integer array	Grid-dependent flag specifying how the sea surface temperature variable is to be acquired or computed for a grid. Values of 0, 1, or 2 are applicable for this flag, and have the same meaning as for ITOPTFLG described above. In contrast to the terrain height data controlled by ITOPTFLG, however, the strict compatibility between coarse and fine grid values of sea surface temperature is not required, although it would be rather senseless for coarse and fine grid values to be grossly different. Sea surface temperature is a variable defined for each surface grid cell in the model and indicates the temperature of any surface water in that cell. For ISSTFLG set to 2, the model calls subroutine SSTINIT that by default sets the sea surface temperature to the value of namelist variable SEATMP. If for a fine grid ISSTFLG is set to 0, the sea surface temperature values for that grid will by default be interpolated from its parent grid. For ISSTFLG set to 1, values for sea surface temperature are interpolated from standard RAMS SST data files. This latter option is used in conjunction with namelist variables ISSTFN. To override any of the above options, subroutines in the file <i>ruser.f90</i> are called, and there a user may customize the sea surface temperature field as described above for modifying TOPT in subroutine TOPTINIT_USER.
IVEGTLG	Grid-dependent flag specifying how the vegetation type variable is to be acquired or defined for a grid.

integer array	Values of 0, 1, or 2 are applicable for this flag. IVEGTFLG = 0 – (permitted only for a nested grid), values of vegetation type are directly assigned from the local grid cell of the parent grid. This applies to each individual subgrid patch as well. IVEGTFLG = 1 – vegetation class data are obtained from a RAMS-specific database which is defined globally at 30-arc-second intervals of latitude and longitude, and whose filename prefix is specified in namelist variable IVEGTFN. When this option is used, and when NVEGPAT is greater than 1, multiple subgrid land patches are filled from the RAMS database according to their relative prevalence in the grid cell. IVEGTFLG = 2 – vegetation class values are defined in subroutine SFCINIT. In this latter case, the value specified in namelist variable NVGCON is used by default for the entire domain. To override any of the above options, subroutine SFCINIT_USER in the file ruser.f90 is called, and there a user may customize the vegetation type field as described above for modifying TOPT in subroutine TOPTINIT_USER.
ISOILFLG integer array	Flag specifying how soil textural class is to be initialized in RAMS. ISOILFLG = 0 – (only allowed for nested grids), soil class is specified from the local value on its parent grid. ISOILFLG = 1 – soil textural class is read from standard RAMS data files. ISOILFLG = 2 – soil textural class for the grid is filled horizontally homogeneously to the value specified by NSLCON. As for vegetation class values described for IVEGTFLG, soil textural class values may be customized in subroutine SFCINIT_USER in the file ruser.f90.
NDVIFLG integer array	Flag specifying how NDVI (Normalized Difference Vegetation Index) is to be initialized in RAMS. NDVIFLG = 0 – (only allowed for nested grids), NDVI is specified from the local value on its parent grid. NDVIFLG = 1 – NDVI textural class is read from standard RAMS data files. NDVIFLG = 2 – NDVI for the grid is filled horizontally homogeneously to the value specified in "leaf3_init.f90". As for vegetation class values described for IVEGTFLG, NDVI values may be customized in subroutine SFCINIT_USER in the file ruser.f90.
EXTRA NOTES ON SURFACE VARIABLES	There are surface variables that do not have available data on standard RAMS data files. These variables are: soil temperature, soil moisture content, snowcover (or temporary surface water) amount, snowcover temperature, snowcover depth, vegetation temperature, vegetation surface water, canopy air temperature, canopy air vapor mixing ratio, and other similar variables that may be unique to the land surface models. These variables are initialized in a default manner in "leaf3_init.f90" subroutine "sfcinit_nofile". For default initialization of soil moisture and temperature, see SLMSTR and STGOFF. Default initial values of vegetation and canopy air temperatures, and canopy air vapor mixing ratio, are the values for the lowest model atmospheric layer. Snowcover and vegetation surface moisture are initialized to zero by default, although snowcover (ground surface water) is set to a positive value in swamps, bogs, and marshes where such land surface types are read from the standard RAMS files (see IVEGTFLG). Upon history restart or history initialization, these surface fields will be interpolated from pre-existing grids and adjusted if needed to fit with assigned surface classes. Note that any of these initial values as well as NDVI, topography, and SST may be overridden by modifying subroutines in the file "ruser.f90". This subroutine is often used, for example, to insert soil moisture and snowcover observational data.
IUPDNDVI integer	Flag for specifying whether observed NDVI values are to be held constant in time during a model run (IUPDNDVI = 0) or are to be continually linearly interpolated in time between the observing times immediately before and after the current model runtime (IUPDNDVI = 1).
IUPDSST integer	Flag for specifying whether observed sea surface temperatures are to be held constant in time during a model run (IUPDSST = 0) or are to be continually linearly interpolated in time between the observing times immediately before and after the current model runtime (IUPDSST = 1). Or IUPDSST=2 to update values during run from 0D FORCINGFILE (see below0).
ITOPTFN char array	Grid-dependent character variable specifying the name of the terrain height data <i>header</i> file, with path names if applicable, to be read in conjunction with namelist variable ITOPTFLG set to 1. These data files are named according to a convention that identifies the latitude and longitude of their southwest corners. The standard files prepared for RAMS each comprise an area of 5 degrees of longitude by 5 degrees of latitude with values defined at 30-arc-second intervals of latitude and longitude and cover all land areas of the earth. These files have names such as EL35N115W, indicating a southwest corner at 35 degrees north latitude and 115 degrees west longitude, with the characters 'EL' serving as a filename prefix for all files in the dataset. The value specified for ITOPTFN would be 'EL' in this case, as the model supplies the remainder

	of the file name according to the file(s) it needs to read. These terrain height files are written in HDF5 format. Sometimes in model applications, fine grids are located within the bounds of a special limited-area terrain height dataset, such as the one described above, and can obtain their terrain height information from that dataset, while coarser grids extend outside the high-resolution dataset, and utilize a coarser-resolution dataset of global coverage, not needing the high-resolution data. This is why ITOPTFN is made grid-dependent.
ISSTFN char array	Grid-dependent character variable indicating the filename prefix (with path names if applicable) of the header file for the sea surface temperature data for initializing each grid. It is similar to namelist variable ITOPTFN .
IVEGTFN char array	Grid-dependent character variable indicating the name (with path names if applicable) of the header file for the vegetation type data for initializing each grid. It is similar to namelist variable ITOPTFN .
ISOILFN char array	Grid-dependent character variable indicating the name (with path names if applicable) of the header file for the soil textural class data for initializing each grid. It is similar to namelist variable ITOPTFN .
NDVIFN char array	Grid-dependent character variable indicating the name (with path names if applicable) of the header file for the NDVI data for initializing each grid. It is similar to namelist variable ITOPTFN .
ITOPSFLG real array	Flag that controls the type of processing of topographic data from input files that are specified in ITOPTFN to final values defined on a RAMS grid. This is a 3-step process that involves topography information being defined successively on 4 different grids, which we call the 'O', 'P', 'Q', and 'R' grids. First, a horizontal interpolation is carried out in order to transfer data from the 'observed' or O grid of the input file, to a polar stereographic grid of comparable resolution, which is our P grid. The P grid uses the same projection as the RAMS grid (the R grid) where the data will reside in its final state but is usually of much higher resolution. Second, data is averaged from this P grid to a lower-resolution Q grid, which is also polar stereographic and has a horizontal grid spacing that is an integer multiple of that on the P grid. This step automatically filters out small scale variations which are not desired on the model grid (see TOPENH). In this second averaging step, a choice of averaging algorithms exists and ITOPSFLG is the flag that selects the choice to be used. ITOPSFLG = 0 – a conventional mean is used where terrain heights for all P grid cells in a single Q grid cell are summed and divided by that number of P values, to obtain the value for that Q cell. ITOPSFLG = 1 – both the conventional mean and a silhouette average are computed, and the value assigned to the Q grid cell is a weighted average of these, with the weights controlled by TOPENH. The silhouette average finds the mean height of the silhouette, as viewed from the east or west, of the set of P grid terrain heights contained within a single Q grid cell, and a separate silhouette height as viewed from the north or south and averages the two silhouette heights together. This becomes the computed silhouette height for that coarse-grid cell. While the conventional average preserves total terrain volume above sea level, the silhouette average adds mass by filling in valleys. It is used to maintain the effective mean barrier height that air must rise to when crossing a topographic barrier such as a ridge. The conventional average lowers this barrier height, particularly when the barrier height is poorly resolved. ITOPSFLG = 2 – an envelope topography scheme is used to obtain Q grid values from P grid values. This scheme is an alternative method of attempting to preserve barrier heights. ITOPSFLG = 3 – a reflected envelope topography scheme is used which aims to preserve both barrier heights and valley depths. Naturally, this method leads to the steepest topography in RAMS, while still filtering the shortest wavelengths. In the third and final step, topography is interpolated from the Q grid to the R grid, and the R grid is usually of moderately higher resolution than the Q grid (see TOPTWVL).
TOPENH real array	Grid-dependent variable specifying the weight given to the <i>silhouette</i> average of terrain height in topographic initialization when ITOPSFLG = 1 (see above), or an envelope orography enhancement factor when ITOPSFLG = 2 or 3. (Also see TOPTWVL .)
TOPTWVL real array	Grid-dependent variable specifying the wavelength, in grid-cell size units, of the smallest horizontal modes of terrain height data which are to be present on a given model grid. It is applicable only for namelist variable ITOPTFLG set to 1. Referring to the description of namelist variable ITOPSFLG above, the value of TOPTWVL controls the ratio of resolution between the Q and R grids. The shortest mode which any grid can resolve is that with a wavelength of twice the grid cell size. In general, the Q grid will contain all wavelengths of topographic data from its own 2 deltax scale and larger. Hence, if the RAMS 'R' grid, to which data is interpolated from the Q grid, has, for example, half the cell size of the model grid, the smallest mode that it will receive from the Q grid will be 4 deltax on the R grid. This smallest mode, in deltax units of the R grid, is the number specified for TOPTWVL. In other words, while the R grid spacing is set by the user (see DELTAX), the Q grid spacing will be TOPTWVL / 2 times the R grid spacing. This is how smoothing of topographic data is achieved in RAMS while allowing the variety of enhancing schemes described for ITOPSFLG. Because the numerical model does not properly handle the smallest

	modes resolvable on a grid, it is generally important not to force these modes into the meteorological fields through overly fine terrain height modes.
IZOFLG integer array	A flag that controls how surface roughness length is computed for computing surface vertical fluxes of momentum, sensible heat, and latent heat. IZOFLG = 0 – the standard roughness height evaluation of soil, snowcover, and/or vegetation surfaces is computed in the land surface submodel of RAMS. IZOFLG = 1 – roughness height is computed from the roughness of subgrid scale topography and may reach several meters. An upper limit to roughness height may be imposed by the user (see ZOMAX). The influence of rough topography on vertical turbulent transport is most properly applied through several (lower) model layers, as well as the surface itself, but in the current implementation, its effect is concentrated solely on the surface fluxes. This tends to overestimate fluxes from the soil and vegetation to the lowest atmospheric level, and this option in RAMS should be used with caution (also see ZOFACT).
ZOMAX real array	Upper bound (in meters) imposed on surface roughness height that is evaluated from subgrid topography (see IZOFLG and ZOMAX).
ZOFACT real array	Roughness factor used in computing surface roughness height from subgrid topography (see IZOFLG and ZOMAX).

\$MODEL_OPTIONS Namelist

The **\$MODEL_OPTIONS** namelist is where the majority of choices for specifying model parameterization options are.

Variable name	Description
ICORFLG integer	Flag specifying whether the Coriolis force is to be activated, and, if the simulation is 2-D, whether the v-component of velocity is to be activated (0 = no, 1 = yes). In a 2-D simulation where only the u and w velocity components are in the grid plane, since the v-component would be uncoupled from u and w without Coriolis force, a value of 0 for ICORFLG deactivates prognosis or use of the v-component. It should be noted that in a 2-D simulation, which assumes that prognostic fields are homogeneous in the y-direction, many factors that control the v-component of wind are excluded from the equation set, and v can become unrealistic, especially where significant terrain exists in the model domain. Thus, 2d simulations should usually exclude consideration of the v-component altogether and use ICORFLG = 0. If the simulation is horizontally homogeneous (namelist variable INITIAL, set to 1), no horizontal pressure gradients are initially available to balance the Coriolis force, and it is useful to compute the Coriolis force only from the difference between the current model winds and a reference state wind (normally the initial model wind interpolated from the sounding). Then, the true horizontal distribution of pressure would be regarded as the sum of the prognosed model pressure and a gradient in geostrophic balance with the reference state wind.
IUGFORCE integer	Geostrophic wind forcing. 0=off, 1=on using input sounding winds, 2=on using time-varying winds from FORCINGFILE. Turn on only if INIITAL=1 or restarting from INITIAL=1 run. Highly recommend if ICORFLG=1 with INITIAL=1.
FORCINGFILE char array	Path to the forcing file for IUGFORCE=2 and IUPDSST=2. Could easily modify source code to include other forcing datasets.
DIVLS real	Large-scale divergence rate for calculating imposed subsidence. A subsidence profile is derived from the DIVLS value where the subsidence at the surface is always 0 m/s.
IBND integer	Flag controlling the lateral boundary conditions (in the x-direction) that are applied on only the coarse grid of a simulation. IBND = 1 – this condition applies only to the normal velocity component (i.e., U at the east and west boundaries), and activates the Klemp-Wilhelmsen condition in which the normal velocity component specified at the lateral boundary is effectively advected from the interior assuming a propagation speed (intended to be similar to a dominant gravity wave phase speed). This phase speed is specified in the namelist in the variable CPHAS. This boundary condition is intended to allow most disturbances to propagate out of the model domain without strongly reflecting back into the interior. IBND = 2 – sets up a cyclic boundary condition between the two x-direction boundaries wherein all prognostic variables in the model, not just the normal velocity component (u), are assigned at the boundaries from values taken from the model interior just inside the opposite boundary. This makes the model periodic in the x-direction, effectively eliminating lateral boundaries from the solution. The number of grid points in the x-direction comprising one complete cycle in the periodic field is 3 less than NNXP for the coarse grid, if the second-order advection scheme is used (Note that we only have 2nd order advection at this time.)
JBND integer	Similar to IBND , but applies to the y-direction, rather than x-direction boundaries. In a 2-D simulation, JBND is irrelevant.

ISPONGE_PTS Integer array	Grid dependent, number of grid points in the lateral boundary sponge zone for nudging of variables from the parent grid to the next nested grid. Useful default (minimum) value is 5 points but may be increased if desired for high resolution (~LES) nests. Recommended to use $\leq \sim 2 \times$ the number of grid points as the nest ratio (e.g., for a 4-1 nest grid spacing ratio, recommend ISPONGE_PTS ≤ 8 , as nudging is performed to the boundary value of the parent grid only, not to the full underlying parent grid structure. Values cannot be larger than the number of columns or rows in a subdomain for parallel decomposition. Example, for 3 grids (ISPONGE_PTS =0,5,5,) where first value is always zero for parent grid.
SPONGE_TAU real array	Grid dependent, user-defined lateral boundary nested-grid sponge zone nudging time scale (seconds) for having a parent grid effectively nudge features into the boundary zone of a nested grid. Values between 10 s and 60 s have been tested. Larger values (slower timescales) can result in more delayed passage of features from the parent to the nest.
CPHAS real	Used in conjunction with IBND and/or JBND set to 1 and specifies the characteristic propagation speed of internal gravity waves used in the Klemp-Wilhelmsen lateral boundary condition. A value of 20 m/s is a common setting for this parameter. Use of larger values, even very much larger, still allows propagation of gravity waves out of the domain reasonably well, whereas much smaller values tend to lead to stronger reflection of the waves back into the domain interior. A very large value of CPHAS is equivalent to the zero-gradient boundary condition. Note that linear noise features that are perpendicular to boundaries are often the results of wave issues and/or diffusion. Using higher values of CPHAS is known to reduce or eliminate this obvious noise that appears in fields of W and/or condensate.
LSFLG integer	Complements IBND and JBND as a lateral boundary condition flag, applying to all variables other than the normal velocity component. It deals with all these remaining variables in the same way. LSFLG applies only for IBND = 1 and for JBND = 1. LSFLG = 0 – sets the lateral boundary value of each variable to the value in the field immediately adjacent to the boundary in the interior. This is a zero-gradient condition. This option is the most commonly used for horizontally homogeneous simulations (the INITIAL variable, set to 1), and is recommended as a first-try option. Occasionally at inflow boundaries, this zero-gradient option has been found to cause a runaway trend to a field variable and must then be replaced by another option. LSFLG = 1 – is very similar to the 0 value, using a zero-gradient condition at inflow boundaries, but setting values at outflow boundaries such that the second spatial derivative in the direction normal to the boundary is zero. This is simply an extrapolation from two interior points. This condition is superior for some cases to the simpler zero-gradient option at the outflow boundary, but usually makes no significant difference. LSFLG = 2 – identical with the value of 1 for the outflow boundary condition but holds variables constant in time at inflow boundaries. This is a possible remedy to the runaway problem sometimes encountered at inflow boundaries, as described above, but is not without its own shortcomings. In cases where flow reverses in time at a lateral boundary, alternating between inflow and outflow, option 2 for LSFLG will cause the boundary condition to alternate between constant in time and time-variant. This may generate an undesirable model response in some cases. Another problem with holding the variables constant in time at an inflow boundary is that those constant values are continuously advected into the model interior and may become incompatible with the evolving situation in the domain. For example, if a boundary layer is growing in the model in response to solar heating, the constant-in-time inflow condition will cause the initial cooler values of temperature to flow into the model from the boundary, artificially cooling the boundary layer. A good solution for this situation is to directly specify an appropriate temporal evolution of the boundary value, still using the LSFLG = 2 option. This is not presently a standard feature in the code but can easily be implemented by the user in subroutine LATSET, located in the file <i>rbnd.f90</i> . LSFLG = 3 – leaves values at all boundaries, inflow and outflow, unchanged in time. This option should be used only for variable initialization simulations (INITIAL = 2), and has sometimes, but not always, been found to give better results than LSFLG = 0 for INITIAL = 2.
NFPT integer	Number of grid levels on the coarse grid, starting from the top of the model domain, in which the Rayleigh friction absorbing layer will be activated (also see DISTIM). A value of 0 leaves Rayleigh friction inactive. Rayleigh friction must be applied only in stable air, and thus requires the model domain to extend sufficiently high, most commonly in the stratosphere. The bottom layer of the Rayleigh friction zone should be above the top of whatever meteorological system is important to the simulation, so as not to directly interfere with its behavior. This normally requires adding more model levels in the vertical than would otherwise be used in a simulation not employing Rayleigh friction. Rayleigh friction is a relaxation of all three velocity components and potential temperature toward the undisturbed, horizontally homogeneous reference state values. Its purpose is to damp gravity wave and other disturbances which approach the top model boundary, so that they will not be reflected back downward. Because prognostic fields which are not initialized horizontally homogeneously cannot sensibly be forced toward a horizontally homogeneous state, Rayleigh friction must not be used with namelist variable INITIAL set to 2. However, top boundary nudging (which damps gravity waves) for horizontally inhomogeneous initialization may be activated elsewhere (see TNUDTOP and ZNUDTOP .)
DISTIM real	Used in conjunction with namelist variable NFPT in activating Rayleigh friction in the upper region of the model. DISTIM specifies a dissipation time scale in seconds, representing the e-folding decay time of a disturbance being damped by the Rayleigh friction layer in the absence of any other source or sink terms. This decay rate strictly applies at the top boundary of the model only, and the strength of the damping

	decreases linearly with height, reaching zero at the lowest of the NFPT levels in which the damping is activated. DISTIM should be related to the rate at which the model is generating upward propagating gravity waves, and normally ranges from 60 to 200 seconds. DISTIM must be at least twice the value of DTLONG for numerical stability. The model solution proves to be rather insensitive to DISTIM , provided it is in or near this range.
ISWRTYP integer	Flag specifying options for evaluating shortwave radiative transfer in the model (0 = no radiation, 1 = Chen and Cotton parameterization, 2 = Mahrer and Pielke parameterization, 3 = two-stream parameterization developed by Harrington, 4 = BUGSRAD, 5 = RTE+RRTMGP). The Mahrer and Pielke scheme is the simplest and by far the least expensive computationally. The main reason for this is that it ignores liquid and ice in the atmosphere, although it does account for water vapor. This scheme should not be used, for example, if attenuation of solar radiation by clouds is important to the simulation. The Chen and Cotton scheme does account for condensate in the atmosphere, but not whether it is cloud water, rain, or ice. This is a major limitation. The Harrington parameterization accounts for each form of condensate (cloud water, rain, pristine ice, snow, aggregates, graupel, and hail) as well as water vapor, and even utilizes information on ice crystal habit. In addition, the scheme adds upper atmospheric levels for radiation computation for cases where the model domain does not extend up to at least 25 km (roughly the height of the ionosphere). This is the most sophisticated parameterization and is recommended over the Chen and Cotton scheme. It is, however, still the most computationally expensive scheme, although work is planned to improve its performance. (See also ILWRTYP and RADFRQ.) The BUGSRAD scheme does not consider aerosol and only considers the cloud and cloud-ice hydrometeor species. The RTE+RRTMGP scheme is considered the best and most modern scheme. It considers all aerosol and hydrometeor categories from the bulk microphysics scheme.
ILWRTYP integer	Similar to ISWRTYP , but applies to longwave, rather than shortwave, radiation. The same values for ILWRTYP apply with the same meaning as for ISWRTYP .
RADFRQ real	Specifies how often during a model run the radiative parameterization is to be exercised to compute updated values for the radiative contribution to tendencies of both atmospheric and land surface temperatures. The tendencies themselves are applied every timestep of the model integration, but the tendencies are updated only at time intervals specified by RADFRQ . Because computing the radiative contribution to temperature tendencies is a relatively computationally expensive process, it is suggested that this computation be done only at intervals of 600 to 1200 seconds (by specifying such values for RADFRQ). In most situations, this is often enough to account for changes in radiative transfer.
LONRAD integer	Flag specifying whether the longitudinal variation of solar hour angle is to be accounted for in the computation of shortwave solar radiation, or whether these angles are to be assumed constant over the model domain. A value of 0 for LONRAD assumes the hour angle is longitudinally constant, and a value of 1 accounts for its longitudinal variation. The latter choice is the one normally made in a simulation of a real event, while the former choice is used when a certain level of idealism is required in a simulation, such as horizontal homogeneity of external forcing.
RRTMFILE character	Is a single-valued character variable giving the path to the RTE+RRTMGP input radiation data files. By default, this directory of files is included in the "etc" directory of the model release. If you change the name or location of this file, you need to make the appropriate change here in this flag.
NNQPARM integer array	Grid-dependent flag used to select the convective parameterization is to be activated or not. NNQPARM = 0 – no convective parameterization for the grid. NNQPARM = 1 – the Kuo-type parameterization will be run for the grid. NNQPARM = 2 – the Kain-Fritsch parameterization will be run for the grid (requires LEVEL=3 microphysics be active). Convective parameterization is used to vertically redistribute heat and moisture (as if by convection) in a grid column when the model generates a region which is superadiabatic or convectively unstable and when the horizontal grid resolution is too coarse for the model to develop its own convective circulation. Ideally, resolving a convective circulation would require at least a few grid cells to horizontally span an updraft, which for deep convection would normally require the horizontal cell size to be less than 1 or 2 kilometers. Coarser resolution than this would make realization of sufficiently strong vertical motion difficult or impossible to adequately bring about the required vertical exchange of heat and moisture so as to convert the convective available potential energy into other forms. Thus, it is on coarser grids where a parameterized convective adjustment becomes necessary. Unfortunately, the convective parameterization schemes currently available assume the grid size in the horizontal to be around 20 kilometers or greater. This means that convective parameterization may be activated on any grid of this resolution, but that at resolutions between about 2 and 20 kilometers, no adequate convective adjustment scheme exists.
CONFRQ	Time interval in seconds during a simulation at which contributions to atmospheric tendencies in temperature and moisture by convective parameterization are to be computed. These tendencies are applied to the

real	temperature and moisture fields every timestep of the model integration but are updated only at the time interval specified for CONFRQ. There are two reasons why those contributions should be updated less frequently than every timestep. One is that a computational expense is involved in exercising the convective parameterization, and frequent exercise thus slows down the model. The change in the tendencies would not be large from one timestep to the next anyway. A second reason has an important physical basis. For example, consider a case of a squall line moving through a grid volume. As the squall line enters the grid volume, it will begin to stabilize the air that it has moved through. However, the air in front of the squall line is still convectively unstable and will contribute to the maintenance of the squall line. The mean atmosphere <i>i.e.</i>, that represented on a coarse grid volume, soon may be stabilized by the convection. If the convective tendencies were to be re-evaluated in the model each 5 minutes or so, the result for the above example would be that the original convection might be shut down in its early stages. A more accurate representation of the squall line, however, would be to let it run through the entire grid volume. This is achieved by setting CONFRQ to a time scale comparable to the length of that life cycle, such as 1200 to 1800 seconds. CONFRQ is relevant only if the parameter NNQPARM is set to 1 for a particular grid.
WCLDBS real	Use in conjunction with NNQPARM and CONFRQ . Specifies the threshold value of vertical velocity in m/s in the model at the diagnosed level of cloud base required to initiate convection. The more positive this parameter, the more difficult is the initiation of convection. Values for this parameter should normally range from about -0.1 to 0.2. This parameter is only relevant for the Kuo scheme.
IKPP integer	Flag to turn on KPP ocean mixed layer model: 0=off, 1=on, 2=on with more diagnostic variables output that are mostly useful for debugging.
DMAXKPP real	Maximum ocean depth in meters.
NKPPZ integer	Number of ocean levels (layers = levels – 1).
DSCALEKPP real	Set exponential scale for stretched ocean depth vertical grid.
FRQKPP real	Timestep for KPP ocean model (seconds). Cannot be less than DTLONG for the coarsest grid. If less than DTLONG, the value is set to DTLONG. Uses same timestep for all grids.
RELAX_SST real	Timescale (days) for SST nudging. Will nudge to initial conditions unless you have updated SST data.
RELAX_OCNT real	Timescale (days) for ocean column temperature nudging. Will nudge to initial conditions unless you have updated ocean column data.
RELAX_SAL real	Timescale (days) for ocean column salinity nudging. Will nudge to initial conditions unless you have updated ocean column data.
KPPITERMAX integer	Maximum number of iterations allowed to compute mixed layer depth.
KPPRNT integer	Flag to turn on sample KPP model output to screen/standard-output. 0=off, 1=on.
UBMN_KPP real	Minimum wind speed (m/s) for computing U^* , T^* , R^* , for KPP wind stress and LEAF-3/SiB surface fluxes. Default is ~1 m/s. RCE ocean simulations have used 4m/s to get better latent heat fluxes.
NPATCH integer	Number of subgrid patches used in the land surface submodel for energy and moisture in soil, snowcover, vegetation, canopy air, and for surface fluxes from these and from water surface areas. NPATCH must be at least 2, to allow for water surface areas and at least one land surface area in each grid cell. Larger values of NPATCH will allow more land surface areas, each with a different vegetation type, soil textural class, and/or wetness index. A value of 2 is often used where the extra detail is not required, while larger values are used for greater accuracy where multiple landuse types may coexist in the same grid cell or where it is desired to model temperature and moisture content of individual, varied ecosystems within the same grid cell.
NVEGPAT integer	Used for any grid whose landuse type is initialized from a standard RAMS file, <i>i.e.</i> , for which IVEGTFLG = 1. For any surface grid cell, especially one covering a relatively large geographic area, many landuse vegetation types will often occur in that area and will be read in from the file. RAMS determines the fractional area of the surface grid cell that is covered by water (oceans, lakes, rivers) and divides the land portion of the grid cell among the N most frequently occurring landuse types read from the file for the cell area, where N is the value specified for NVEGPAT . NVEGPAT should thus usually be set to NPATCH -1 in order to allow for the 1 water patch and fill all land patches for the data file. For special applications, one may wish to set

	NVEGPAT to a value less than NPATCH-1 to reserve additional land patches for special use. Then, the model will still fill the water patch and the NVEGPAT land patches such that they collectively comprise the entire grid cell area. The user must then reduce the fractional areas of 1 or more of the patches to allow the desired fractional area of the special patch(es) whose number is NPATCH-NVEGPAT-1. NVEGPAT may not be larger than NPATCH-1.
ISFCL integer	Flag specifying which of 2 possible options for obtaining surface fluxes of heat and moisture to or from the atmosphere will be used in a simulation. ISFCL = 0 – the user specifies values for namelist variables DTHCON and DRTCON representing the result of subtracting a fictitious ground temperature and moisture mixing ratio, respectively, from the surface atmospheric values of the two variables. These differences are then converted to vertical fluxes of heat and moisture to or from the lowest atmospheric level. This is the simpler of the 2 choices and ignores the soil/vegetation parameterization available in the model. ISFCL = 1 – activates the LEAF submodel. LEAF prognoses soil, vegetation, snowcover, and canopy air temperature and moisture based on vertical diffusion and exchange with the atmosphere of both quantities. Use of LEAF requires radiation be turned on. (Also see NZG, PCTLCON, NSLCON, SLZ, and STGOFF.) ISFCL = 2 – activates the LEAF submodel for water patches but uses the Simple Biosphere (SiB) submodel for land patches. This requires setting initial CO2 profile in CO2_INIT or use of custom CO2 input. ISFCL = 3 – Use this when you want to specify everything about the surface simply: (a) momentum fluxes are hard-coded to zero, (b) latent and sensible heat fluxes are given in W/m2 by DRTCON & DTHCON, respectively, (c) surface albedo is specified by ALBEDO, (d) surface temperature is specified by SEATMP and is used only for calculating RLONGUP. When using this option, only the TOPT file will be made. The surface, NDVI, and SST files are unnecessary. Also, using this scheme, patch properties such as USTAR are not calculated. As such, the aerosol deposition scheme cannot be run.
IFREESLIP integer	Flag for setting freeslip surface boundary conditions. Sets all surface and momentum fluxes to zero. IFREESLIP=0 means normal surface boundary conditions. IFREESLIP=1 means use freeslip conditions.
SIBFILE character	Is a single-valued character variable giving the name (and path, if any) of the SiB surface model land surface morphology tables. The file name is hardcoded as SiB-Veg-Morphology.txt. By default, this file is included in the "etc" directory of the model release. If you change the name or location of this file, you need to make the appropriate change here in this flag.
CO2_INIT real array	CO2 initial vertical profile for SiB submodel. Number of input values should match number of vertical levels from surface to model top. If it does not, it will assign the highest elevation value to the remaining model levels aloft. Values should be in units of parts-per-million (ppm). This could be overwritten with custom observations.
ISOILDAT integer	Flag to tell the model to ingest real soil moisture and soil temperature from the NARR or GDAS-FNL data. ISOILDAT=0 for NO and ISOILDAT=1 for YES to ingest soil data. Currently only these 2 datasets have extracted soil and snow data. Contact Stephen Saleeby (Stephen.Saleeby@colostate.edu for access to the dprep files containing this soil and snow data. If the dprep files have soil and snow data, then the varfiles will automatically contain these fields. The user just has to choose whether to use default soil fields at initialization or the observed fields from the varfiles. Note that soil fields are only applied at initialization (time=0) for RUNTYPE='INITIAL' run using varfiles for initialization (INITIAL=2). Only grids with varfiles can apply observed soil fields to the grid. Also, soil ingest is not done if IPAST_SFC=1 which uses recycled surface fields.
SNUDCENT real array	Soil moisture nudging time scale (seconds) for domain interior. Soil temperature is updated to be in balance with soil moisture as part of the nudging process. This only works if you have soil moisture and soil temperature data in your varfiles which would have had to come from gridded reanalysis for RAMS history files for history file initialization and nudging. Values for this variable would be similar to the central nudging timescale (TNUDCENT).
ISNOWDAT integer	Flag to tell the model to ingest real snow depth and snow mass from the NARR or GDAS-FNL data. ISNOWDAT=0 for NO and ISNOWDAT=1 for YES to ingest snow data. Currently only these 2 datasets have extracted soil and snow data. Contact Stephen Saleeby (Stephen.Saleeby@colostate.edu for access to the dprep files containing this soil and snow data. If the dprep files have soil and snow data, then the varfiles will automatically contain these fields. The user just has to choose whether to use default snow fields at initialization or the observed fields from the varfiles. Note that snow fields are only applied at initialization (time=0) for RUNTYPE='INITIAL' run using varfiles for initialization (INITIAL=2). Only grids with varfiles can apply observed snow to the grid. Also, soil and snow ingest is not done if IPAST_SFC=1 which uses recycled surface fields.
NVGCON integer	Flag specifying the landuse class to be used in the land surface submodel of RAMS, for the case where a simple horizontally homogeneous initialization of landuse type is used (see IVEGTFLG). The landuse class is mostly characterized by its vegetation type, but landuse class also defines whether the surface is water, ice cap, wetland, or unvegetated land. By default, the value of this flag is used to fill all surface grid cells. For horizontally inhomogeneous initialization of landuse type, data are normally input from a standard dataset (see IVEGTFLG). For direct customizing of inhomogeneous landuse type, the default homogeneous value specified here may be overridden in the file <i>ruser.f90</i>. Values for NVGCON ranging from 0 to 21 have

	the following definitions in the standard RAMS code, but new classes are easily added where information is available: <table> <tr> <td>0 Ocean</td><td>11 Tundra</td></tr> <tr> <td>1 Lakes, rivers, streams</td><td>12 Evergreen shrub</td></tr> <tr> <td>2 Ice cap/glacier</td><td>13 Deciduous shrub</td></tr> <tr> <td>3 Desert, bare soil</td><td>14 Mixed woodland</td></tr> <tr> <td>4 Evergreen needleleaf tree</td><td>15 Crop/mixed farming, C3 grassland</td></tr> <tr> <td>5 Deciduous needleleaf tree</td><td>16 Irrigated crops</td></tr> <tr> <td>6 Deciduous broadleaf tree</td><td>17 Bog or marsh</td></tr> <tr> <td>7 Evergreen broadleaf tree</td><td>18 Wooded grassland</td></tr> <tr> <td>8 Short grass</td><td>19 Urban and built up</td></tr> <tr> <td>9 Tall grass</td><td>20 Wetland evergreen broadleaf tree</td></tr> <tr> <td>10 Semi-desert</td><td>21 Very urban</td></tr> </table> Each of these landuse classes is assigned a set of land surface parameter (LSP) values, including leaf area index, vegetation fractional coverage, vegetation height, albedo, and root depth. The LSP values are defined in a data statement in subroutine SFCDATA in the file "leaf3_init.f90".	0 Ocean	11 Tundra	1 Lakes, rivers, streams	12 Evergreen shrub	2 Ice cap/glacier	13 Deciduous shrub	3 Desert, bare soil	14 Mixed woodland	4 Evergreen needleleaf tree	15 Crop/mixed farming, C3 grassland	5 Deciduous needleleaf tree	16 Irrigated crops	6 Deciduous broadleaf tree	17 Bog or marsh	7 Evergreen broadleaf tree	18 Wooded grassland	8 Short grass	19 Urban and built up	9 Tall grass	20 Wetland evergreen broadleaf tree	10 Semi-desert	21 Very urban
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PCTLCON real	Fraction of a surface grid cell, ranging from 0 to 1, that is covered by land surface, implying that the remainder is covered by water. PCTLCON is the default value used when IVEGTFLG is set to 2 (see above). If IVEGTFLG is 0 or 1, PCTLCON is ignored. The distinction between land and water surfaces in the model influences the values of surface fluxes of momentum, heat, and moisture that are exchanged between the atmosphere and surface. Separate calculations of these fluxes are made for each grid cell for the soil and water surfaces, and the resultant net flux is weighted between these two according to the relative area covered by each type of surface. By default, the value assigned to PCTLCON is assigned to each surface grid cell in the model. If, as is usually the case, the user wishes to define a horizontally variable distribution of land percent values, subroutines in the file "ruser.f90" need to be modified.																						
NSLCON integer	Soil textural class used in the land surface submodel of RAMS. NSLCON is the default value used when ISOILFLG is set to 2 (see above). If ISOILFLG is set to 0 or 1, PCTLCON is ignored. The soil type controls important soil properties that ultimately affect the influence by the soil on the atmosphere. These properties are the thermal diffusivity, specific heat, moisture capacity (porosity), and the diffusivity of moisture. The thermal properties are themselves highly influenced by the moisture content of the soil, but the particular relation between them is dependent on the soil class. A total of 12 different soil textural classes are parameterized in the model. Below is a list of each soil class and its value of NSLCON: <table> <tr> <td>1 Sand</td><td>7 Silty clay loam</td></tr> <tr> <td>2 Loamy sand</td><td>8 Slay loam</td></tr> <tr> <td>3 Sandy loam</td><td>9 Sandy clay</td></tr> <tr> <td>4 Silt loam</td><td>10 Silty clay</td></tr> <tr> <td>5 Loam</td><td>11 Clay</td></tr> <tr> <td>6 Sandy clay loam</td><td>12 Peat</td></tr> </table> As in the case for PCTLCON, the value assigned to NSLCON in the namelist is in turn assigned to the soil type for each surface grid cell in the model. If it is desired to make the soil type horizontally inhomogeneous, the user may modify subroutines in the file "ruser.f90".	1 Sand	7 Silty clay loam	2 Loamy sand	8 Slay loam	3 Sandy loam	9 Sandy clay	4 Silt loam	10 Silty clay	5 Loam	11 Clay	6 Sandy clay loam	12 Peat										
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5 Loam	11 Clay																						
6 Sandy clay loam	12 Peat																						
ZTROUGH ZMROUGH real	ZTROUGH is used for all scalars. ZMROUGH for winds. Roughness length, a parameter important in representing the effects of ground roughness on the surface fluxes of momentum, heat, and moisture. The parameterization of these fluxes is designed according to surface similarity theory, in which the roughness length is a crucial parameter. A roughness length of 0.01 meter or less represents a smooth surface, while a roughness length of 1.0 meter represents a very rough surface such as a forest of tall trees. As in the case of PCTLCON, the value assigned to ZROUGH in the namelist is in turn assigned to each grid point in the model. Should the user wish to make roughness length a location-dependent parameter, subroutines in the file "ruser.f90" must be modified. ZROUGH only refers to the soil areas of the model surface; the vegetation roughness lengths vary with type and are set in subroutine SFCDATA. These values are only used for soil patches when ISFCL=0 or for the water patch if ICHARNOCK=0. The LEAF and SiB schemes continue to use only one value of roughness.																						
ICHARNOCK integer	Flag to sue the Charnock ocean roughness parameterization. If set to zero, instead use ZTROUGH and ZMROUGH for the ocean patch.																						
ALBEDO real	Albedo or reflectivity of the ground surface to solar radiation. This parameter only applies when the surface submodel is not activated, <i>i.e.</i>, for ISFCL set to 0. In this case, the albedo is not involved in the computation of surface sensible and latent heat fluxes; it is only used to compute solar radiation reflected upward from the ground, which is needed in the atmospheric radiation budget.																						
SEATMP	Temperature in Kelvins of the surface water temperature. SEATMP is the default value used only if																						

real	ISSTFLG is set to 2. If ISSTFLG is 0 or 1, SEATMP is ignored. Horizontally homogeneous initialization of sea surface temperatures by SEATMP may be overridden by modifying subroutine SFCINIT_USER in the file <i>ruser.f90</i> . Surface water temperature is relevant for any surface grid cell which is not completely covered by land (see PCTLCON), and strongly influences the fluxes of heat and moisture between the surface and atmosphere. If ISSTFLG is 2 and SEATMP is therefore used, the model assumes that the surface water temperature is constant in time during a simulation.
DTHCON real	Parameter representing the temperature difference in degrees Kelvin obtained by subtracting a fictitious soil temperature from the temperature of the lowest level of the atmosphere. It is used as a simple means for imposing a surface source or sink of heat to the atmosphere in the case where the soil model is not activated, that is for ISFCL set to 0. For example, setting DTHCON to -10 would impose a flux of heat into the atmosphere equivalent to that which would occur over a ground surface 10 K warmer than the air a height around one half DELTAZ above the ground.
DRTCON real	Similar to DTHCON , being relevant only for ISFCL set to 0, but applies to a difference in water vapor mixing ratio, rather than to a temperature difference. For example, setting DTHCON to -0.002 would cause an upward flux of moisture to the atmosphere equivalent to that which would occur if the equilibrium vapor mixing ratio in the soil were 2 grams per kilogram higher than the ambient vapor mixing ratio at low levels in the atmosphere.
SLZ real array	Depth of the levels in the soil model in meters. There needs to be as many values specified as NZG beginning with the deepest level. The SLZ values pertain to the <i>bottom</i> of the soil layers. Unlike previous RAMS' versions, SLZ(NZG) should not be set to 0.
SLMSTR real array	Used to initialize the moisture content of the soil. The number of values specified must equal the number of soil levels NZG with the values must be separated by commas. The first value specified for SLMSTR applies to the deepest soil level, with the last value representing the topmost layer of soil. Allowable values for this parameter range from 0.0, representing totally dry soil, to 1.0 which represents totally saturated soil. A more realistic lower limit is about .15 since soil cannot lose all of its moisture, even when exposed to direct sunlight for extended periods, and RAMS will in fact impose a minimum value on soil moisture to prevent its initialization too dry. The values represent fraction of total water capacity that the soil can hold. The equilibrium water vapor mixing ratio of the soil is the quantity with which the ambient atmospheric water vapor mixing ratio interacts in determining the flux of moisture between the atmosphere and soil. This soil equilibrium value is a highly nonlinear function of the soil moisture content (represented initially by SLMSTR). The equilibrium vapor pressure is essentially constant for soil moisture below about 0.2 and above about 0.7 [depending on the soil type (see NSLCON)] and varies sharply in between those values. The SLMSTR values are by default used to fill all soil grid points horizontally-homogeneously, but horizontal variations in initial soil moisture may be specified by the user by altering the code in subroutines in the file "ruser.f90". See ISOILDAT for soil and snow data ingest.
STGOFF real array	Used to define a vertical temperature profile in the initial conditions for the soil model and represents a deviation from the temperature in degrees K in the lowest atmospheric level in the model. The values must be equal in number to NZG and must be separated by commas. The first value of STGOFF applies to the deepest level of soil, while the last value applies to the topmost soil level. For example, if the first value of STGOFF is set to 5.0, the deepest soil level will initially have a temperature 5 degrees K. warmer than the initial temperature of the lowest atmospheric level. As for many other soil variables in the namelist, the values specified in STGOFF are by default used to horizontally homogeneously initialize all soil model grid points. A user may impose horizontal variations in the initial soil temperature by modifying the code in subroutines in the file "ruser.f90".
IDIFFK integer array	Grid-dependent flag that controls the type of parameterization to be used for computing both horizontal and vertical diffusion coefficients. Values for this flag are 1-4. A value of 1 or 2 is appropriate to model grids in which the horizontal grid spacing >> vertical spacing such that dominant convective motions are not resolved. Horizontal diffusion in such cases is normally required to be stronger (for numerical damping) than physically justifiable, and the model assumes complete decoupling of the horizontal and vertical diffusion in all aspects, including computation of deformation rates, length scales, and stress tensor components. IDIFFK = 1 or 2 – the horizontal diffusion coefficients are computed as the product of horizontal deformation rate (horizontal gradients of horizontal velocity) and a length scale squared, based on the original Smagorinsky formulation. The length scale is the product of x-direction grid spacing DELTAx and parameter CSX . IDIFFK = 1 – vertical diffusion is parameterized according to the Mellor and Yamada scheme, which employs a prognostic turbulent kinetic energy (TKE). IDIFFK = 2 – vertical diffusion is computed from a 1-dimensional analog of the Smagorinsky scheme in which vertical deformation is evaluated from vertical gradients of horizontal wind, and vertical length scale is the local vertical grid spacing times namelist parameter CSZ . Modifications of the vertical diffusion coefficient due to static stability or instability are used, based on formulations of Hill & Lilly. The Lilly modification is in the form of a multiplying factor, equal to $\sqrt{1-R \text{ kh/km}}$ where R is the Richardson # and kh/km is the ratio of the scalar to momentum vertical diffusion coefficients specified in ZKHKM . The factor is greater than 1 in unstable cases (<i>i.e.</i> , where wind shear and/or unstable lapse rates make the Richardson number sufficiently low and is less than 1 in stable cases. The Hill modification applies only to regions of unstable lapse rates (having negative squared Brunt-Vaisala frequencies) and consists of adding the

	absolute value of the Brunt-Vaisala frequency squared to the deformation rate, to obtain a modified inverse time scale for the diffusion coefficient computation. The Lilly and Hill modifications were originally designed for use without each other, although we have found that the added vertical diffusion in unstable air obtained by using both together is usually desirable. Values of 3,4 for IDIFFK are usually appropriate on grids having comparable horizontal and vertical spacings and are intended for use in situations where dominant 3D motions, such as cumulus or BL convection, are resolved. In these cases, RAMS computes diffusion coefficients from the 3D rate-of-strain tensor and uses the same or similar diffusion coefficients for horizontal and vertical diffusion. IDIFFK = 3 – horizontal and vertical diffusion coefficients are computed as the product of the 3-D rate-of-strain tensor and a length scale squared. The length scale is the product of the vertical grid spacing and namelist parameter CSZ. The Hill and Lilly stability-dependent mods described above also apply to IDIFFK = 3. IDIFFK = 4 – vertical and horizontal diffusion are parameterized according to the Deardorff scheme, which employs prognostic subgrid TKE. This scheme is intended only for the specific purpose of performing a large eddy simulation (LES) in which it is assumed that resolved eddy motions in the model perform most of the eddy transport. The parameterized diffusion only represents the subgrid turbulent mixing. Thus, the Deardorff scheme is not appropriate for horizontal grid cell sizes larger than a couple hundred meters. In both the Deardorff and Mellor-Yamada schemes, the prognostic kinetic energy is generated by means of shear and buoyancy processes, and a parameterized pressure-work term, and is destroyed by potentially all of the above plus a parameterized dissipation term. The kinetic energy is advected and diffused. The result of these processes yields a field of kinetic energy from which the diffusion coefficients are locally diagnosed.
IDIFFPERTS integer	Choice to use diffusion of perturbations rather than diffusion of scalars with regards to diffusing THP and RTP. 0=normal scalar diffusion, 1=diffuse perturbations from BASE state THP,RTP, 2=diffuse perturbations from the current domain MEAN state THP,RTP, 3=diffuse perturbations from the “varfile” state THP,RTP (option 3 meant for case studies where “varfiles” or variable initialization files are available. Using IDIFFPERTS options can help eliminate artificial cloud/stable layers that appears in some circumstances. Default should be IDIFFPERTS=0.
IHORGRAD integer	Flag to control how horizontal gradients are taken throughout the computation of the diffusion terms. IHORGRAD = 1: Normal sigma-z gradients are taken, which consist of combining the gradient along the sloping coordinate surface with a vertical component to approximate a true Cartesian horizontal gradient. This is the preferred setting, as it is a flux-conservative scheme. However, if the topography is too “steep” (see discussion of DELTAZ), the allowable DELTAZ for a given topography may be limited. IHORGRAD = 2: Alternate technique to compute the horizontal gradients by first interpolating the variable vertically to the level at which the gradient is to be taken. This scheme is non-conserving, so it is more applicable for “short” runs. However, it allows a smaller DELTAZ for a given topography.
CSX real array	Grid-dependent parameter serving as a coefficient in computing horizontal diffusion coefficients, in the case where IDIFFK is set to either 1 or 2. It is used in computing the horizontal diffusive length scale, defined as the product of CSX and the x-direction grid spacing. Doug Lilly and others have proposed specific values for CSX based on theory, although the philosophy on which such a theory is based has been a matter of controversy. Setting CSX to 0.135 is effectively the value proposed by Lilly to yield diffusion coefficients of a strength that just provides adequate damping of the smallest resolvable features on the grid (to prevent an unrealistic accumulation of energy at that small scale). However, justification for a considerably larger of CSX of 0.32, has been argued by Paul Mason for more strongly filtering features represented on the grid, so that the only features with wavelengths longer than about 4 or 5 times the grid spacing will exist at any significant amplitude on the grid. This choice is desirable since the the smaller wavelengths of 2 and 3 times the grid spacing are not properly handled by numerical finite differencing schemes. The price paid for this stronger diffusive filter is the requirement of more grid cells for simulating a particular feature. The user must make a judgment here as to how strong to make CSX. This is often done on a case-by-case basis based on a subjective decision as to whether the model prognostic fields are “sufficiently smooth”. Simulations can be very sensitive to CSX and CSZ such that higher or lower values of CSZ, are known, for example, to help eliminate the formation of unrealistic “stable layers” that sometime form in low shear environments.
CSZ real array	Similar to CSX, but it applies to the computation of vertical, rather than horizontal, diffusion coefficients, and is relevant only for IDIFFK set to 2 or 3.
XKHKM real array	Grid-dependent parameter in which the user specifies the ratio of strength of horizontal diffusion coefficients between scalars and velocity to be used in a simulation. It is used to multiply the momentum diffusion coefficient to obtain the scalar coefficient and applies to cases where IDIFFK is set to 1 or 2. Deardorff found that the scalar coefficient had to be around 3 times larger than the momentum coefficient in strongly convective boundary layers for adequate mixing of all fields. For stable situations, the ratio may decrease to 1. It has been our usual practice to set XKHKM to 3, a value which applies throughout the model domain regardless of local stratification. The Deardorff parameterization for large eddy simulations (activated by setting IDIFFK to 4) includes an automatic computation of this ratio for each grid cell in the domain based on the local conditions.
ZKHKM	Similar to XKHKM, but applies to the computation of vertical, rather than horizontal, diffusion coefficients. It

real array	is applicable only in cases where IDIFFK is set to 2 or 3.
AKMIN real array	Multi-valued, grid-dependent parameter used to impose a minimum value on the value of the horizontal diffusion coefficients throughout the model domain. It is only relevant for cases where IDIFFK is set to 1 or 2. The minimum value is sometimes required when the local fluid deformation rate happens to be close to zero in a region, resulting in very weak diffusion there. Setting AKMIN to a value greater than 0 will prevent the horizontal diffusion coefficients everywhere from falling below the minimum. AKMIN is scaled such that if set to 1.0, the minimum value placed on the diffusion coefficients should be about the same as normally obtained from the basic deformation scheme itself (without buoyancy effects) regardless of grid spacing. It is suggested that AKMIN be set to a low value, say 0.1, so that the standard horizontal diffusion parameterization be given a chance to do its job. If it is found that horizontal diffusion is inadequate (for example, as seen by considerable grid noise appearing in a contour plot of model fields), first try increasing CSX up to as much as .32, and if more diffusion is still required, then increase AKMIN to as much as 1 or 2, as required. From experience, low values of AKMIN (ie. 0.2 or so) tend to be appropriate for low shear tropical scenarios, whereas higher values (i.e., 2.0+) tend to work well in high shear mid-latitude supercell thunderstorm type of environments. Higher values help reduce excessive noise but can strongly impact feature evolution and precipitation amounts.
FRACSAT real array	Non-grid-dependent variable used in computing Brunt-Vaisala frequency for the Richardson number calculation in diffusion. For non-cloudy grid cells, the moist-BV (MBV) is computed as in Durran & Klemp (1982, DK82). For cloudy grid cells, saturated-BV (SBV) is used. However, there can be cloudy grid cells that are sub-saturated and undergoing evaporation. In addition to the condition for the presence of cloud water, is it required that: RV (water vapor mixing ratio) > FRACSAT * RVLS (saturated water vapor mixing ratio). FRACSAT near 1.0 keeps SBV from only being applied to grid cells that are cloudy and saturated. FRACSAT < 1.0 allows the use of SBV is sub-saturated conditions. The user can choose how sub-saturated the conditions can be for application of SBV. Experience shows that for non-layered clouds, using FRACSAT from 0.96-0.99 helps limit artificial cloud layers. However, for simulating layered clouds like fog, stratus, or stratocumulus, using a lower value of FRACSAT (< 0.90) is better. The user may need to experiment with this variable depending on the simulation type.
IBUBBLE integer	Warm bubble initialization flag. Used for idealized initialization of thermal and moisture perturbations at time=0. Value of 0 is off and 1 = square bubble, 2 = cosine-squared bubble, 3 = random perturbations. Bubble is initialized in ruser.f90. User MUST customize the bubble.
IBUBGRD integer	Parameter to specify on which grid a bubble is initialized.
IBDXIA integer	X grid point value used to define the western edge of the bubble.
IBDXIZ integer	X grid point value used to define the eastern edge of the bubble.
IBDYJA integer	Y grid point value used to define the southern edge of the bubble.
IBDYJZ integer	Y grid point value used to define the northern edge of the bubble.
IBDZK1 integer	Z grid point value used to define the lower edge of the bubble.
IBDZK2 integer	Z grid point value used to define the upper edge of the bubble.
BTHP real	Theta-IL (THP, ice-liquid potential temperature) perturbation (Kelvin) (THP + BTHP). This is imposed as an absolute change in THP.
B RTP real	Total water mixing ratio (RTP) perturbation (kg/kg) (RTP * (1 + B RTP)). This is imposed at a fractional change as noted by the equation above.
ICONV integer	Parameter to specify implementation of an idealized convergence zone used for generating idealized convection. A value of 0 is off, 1=Gaussian in x and y, U and V, 2=Gaussian in x and y, U only, 3=Gaussian in x and y, V only, 4=Gaussian only in x, U only, 5=Gaussian only in y, V only. See Loftus et al. (2008, Monthly Weather Review) for details. Convergence can be imposed for a particular duration of time or until a certain domain maximum vertical velocity is achieved. See additional flags below.
ICONGR integer	Parameter to specify on which grid to implement idealized convergence forcing.

ICICENT integer	Convergence X-direction (I-direction) grid point center of convergence.
ICJCEN integer	Convergence Y-direction (J-direction) grid point center of convergence.
CXRAD real	Convergence radius (meters) of influence in the X-direction in the Gaussian function.
CYRAD real	Convergence radius (meters) of influence in the Y-direction in the Gaussian function.
ICVERT integer	Convergence vertical structure: 1=Linear decrease with height and maximum at the surface, 2=Elevated 1/2 vertical wavelength
ICKMAX integer	Maximum K (vertical grid point location) for ICVERT=1 (ignored for ICVERT=2)
CZRAD real	Vertical radius (meters) for ICVERT=2 (ignored for ICVERT=1)
ICKCENT integer	K center (vertical grid point location) for ICVERT=2 (ignored for ICVERT=1)
CDIVMAX real	Divergence amplitude (s^{-1}) (negative = convergence). See Loftus et al. (2008) for guidance. Stronger convergence gives greater upward vertical motion more quickly.
CTAU real	Timescale (seconds) over which convergence zone strength increases to amplitude CDIVMAX. Value of 1 gives full convergence strength. Larger values give slower convergence.
CTMAX real	Parameter to set the duration of convergence forcing. If CTMAX < 0, the absolute value of CTMAX is the vertical velocity threshold past which to terminate forcing. If CTMAX > 0, this value is the time (in seconds) past which to terminate forcing.
IAEROLBC Integer array	Setting grid-dependent variable IAEROLBC to 1 will set the aerosol LBCs for horizontal grid rows and columns i=2, j=2, i=nx-1 and j=ny-1 to the outer boundary value. The strength of nudging is modified by a weighting function controlled by namelist variable BCTAU. A value of 0 is off. This will function for zero gradient and radiative boundary conditions only (see LSFLG = 0 or 1). This would typically only be used on an outer domain but could be used for nested grids if needed. Used only for aerosol situations without aerosol sources (i.e., prescribed CCN, dust, or sea-salt).
ICO2LBC Integer array	Same as grid-dependent namelist variable IAEROLBC but used to force CO2 at lateral boundaries when SiB land surface model is being run.
BCTAU real array	The grid dependent flag BCTAU is a number in seconds that is used to compute the inverse time weighting. The time weight gets computed as (timestep/BCTAU) and lets the user control the strength of resetting custom profiles. If BCTAU = timestep then full aerosol resetting occurs; if BCTAU > timestep, the resetting will be more gradual. If BCTAU < timestep it will be forced to be equal to the timestep on the respective grid. Controls weighting related to flags IAEROLBC and ICO2LBC.
IRCE Integer array	Flag for using typical setup for simulation running Radiative Convective Equilibrium (RCE). 0=Off, 1=On.
RCE_SZEN real array	Constant solar zenith angle for RCE if IRCE=1 (Values from 0.0 to 90.0 degrees) (50.5 degrees might be a typical value).
RCE_SOLC real array	Solar constant for RCE if IRCE=1 (650 W/m2 might be a typical value).
RCE_UBMN real array	Constant minimum wind speed used to compute surface fluxes if IRCE=1 (Typical value of 4.0 m/s for RAMS in RCE).
RCE_BUBL real array	Maximum temperature perturbation in Kelvin for initial randomized warm bubbles if IRCE=1 (Typical value for RAMS in RCE is 0.50K).
ITRACER integer	Integer number of tracers to include in a simulation. Initialize in subroutine "init_tracer" within micro/mic_init.f90. Note that tracers (a.k.a. 3D scalars) are allocated on all grids, but initialization does not have to occur on all grids. User sets these up in custom format.

ITRACHIST integer	Flag to indicate if the user wants to initialize tracers on history initialization and/or history restart. 0=NO, 1=YES. A value of 1 will erase all former tracer values and set to new values.
LEVEL integer	Flag that specifies the level of moisture complexity to be activated in the model. This flag is the primary means by which the user tells the model whether to consider the effects of moisture in the simulation, and if so, to what degree. LEVEL = 0 – causes the model to run dry, completely eliminating any process which influences or is influenced by any phase of moisture. With this option, radiation parameterizations (controlled by ISWRTYP and ILWRTYP) must be turned off. LEVEL = 1 – activates advection, diffusion, and surface flux of water, where all water substance in the atmosphere is assumed to occur as vapor even if supersaturation occurs. The value of 1 also activates the buoyancy effect of water vapor in the vertical equation of motion, as well as the radiative effects of water vapor if radiation is activated elsewhere. LEVEL = 2 – activates condensation of water vapor to cloud water wherever supersaturation is attained. The partitioning of the total water substance into vapor and cloud water is purely diagnostic in this case. No other forms of liquid or ice water are considered. Both the positive buoyancy effect of water vapor and the liquid water loading of cloud water are included in the vertical equation of motion. Radiative effects of both water vapor and cloud water are activated if the radiation parameterization is itself activated. LEVEL = 3 – activates the bulk microphysics parameterization, which includes cloud water, drizzle, rain, pristine ice, snow, aggregates, graupel, and hail, or certain subsets of these as specified by ICLOUD, IDRIZ, IRAIN, IPRIS, ISNOW, IAGGR, IGRAUP, and IHAIL. This parameterization includes the precipitation process. LEVEL = 4 – activates the Hebrew University Cloud Model – Spectral Bin Microphysics (HUCM-SBM) parameterization. Please see the HUCM-SBM instructions for setting the microphysics parameters below that are associated with aerosols and hydrometeor types. Some of the remaining namelist flags in this section overlap between LEVEL=3 and LEVEL=4 and some are unique to each. Exercise caution when setting these flags. Uses the microphysics flags IPRIS, IGRAUP, IHAIL, CCN_MAX, CIN_MAX, AERO_MEDRAD, IAERO_CHEM, IIFN, NDTCOLL. ISNOW and IAGGR are not really used, but need to be 0 (along with IPRIS, IGRAUP, IHAIL) if trying to run liquid only.
ISCM integer	Flag to signal for output of variables in single column model (SCM) output format for use with SCM version of the microphysics, radiation, and aerosols. Output for the SCM requires the user to create the default directories named “ scm.in ” and “ scm.out ” (all lower case) within the current runtime directory. Single column text files are written for every allocated physics variable. Keep this set to ISCM=0 except for custom simulations.
ISCMX, ISCMY integer	These indicate the I & J (X & Y) location of the column to output for the SCM.
SCMTIME real	Time in seconds of the runtime at which to output the SCM data column text files.
ICHECKMIC integer	Choices: 0=off, 1=on. Flag for checking microphysics and aerosol variables for NaN values and for negative values of condensate. This check is called after MICRO in subroutine TIMESTEP. Note that INTEL compiler works by default with NaN checking, but for PGI compiler you need to compile with the flag “-Kieee”. Runs for LEVEL ≤3.
IMBUDGET integer	Choice for computing and outputting microphysics budget variables in each analysis file. These are the “Total” budgets which are summed over time between output write times of ANALYSIS files and then reset to zero. Budgets are not reset on the LITE or MEAN output file frequency. IMBUDGET=0 is off. IMBUDGET=1 computes basic budgets such as vapor growth of all liquid species. IMBUDGET=2 computes more specific budgets such as vapor growth of rain drops. IMBUDGET=3 computes additional custom budgets. A list of output “total” budgets can be determined in mem_micro.f90 and the REVU output variables document. Runs regardless of LEVEL , but some output variables are based on LEVEL and allocated microphysics variables.
IRIME integer	Choice of standard bulk riming (IRIME=0) or hybrid-bin riming of cloud droplets via method of moments (IRIME=1). (Saleeby and Cotton 2008, JAMC).
IPLAWS integer	IPLAWS=0 uses original hydrometeor species fall speed power laws from RAMS 4.3. IPLAWS=1 uses non-banded, but newer power laws via Mitchell (1996). IPLAWS=2 uses size banded newer power laws via Mitchell (1996) and modified by Rob Carver and Jerry Harrington (PSU).
ISEDIM integer	ISEDIM=0 uses standard hydrometeor sedimentation with relative displacement for binned fall distance. ISEDIM=1 uses a true-bin-hybrid sedimentation that requires extra lookup table dimensions but improved realism in fall distance.

IKERNELA integer	IKERNELA=1 uses the Long (1974) liquid droplet collection kernel while IKERNELA=2 uses the Hall (1980) kernel.
ICLOUD integer	Values are 0-5. Values of 1 or 4 are normally very preferable over ICLOUD = 2. Cloud droplet number prediction is an option and is used by setting ICLOUD = 5. ICLOUD = 5 nucleates cloud droplets from a concentration of cloud condensation nuclei (CCN) that is specified by the user in CCN_MAX (see below) in units of number per milligram of air. Other categories of aerosol are available as well (see below). Assumed profiles of aerosols are set in mic_init.f90. The default is exponential decrease with height, similar to pressure, with the lowest model level value set to CCN_MAX . Initial concentration in #/mg is roughly equal to #/cm ³ in the lower troposphere but is less than this at higher levels. Aerosols evolve in time after initiation. There is no bulk source function of CCN, but there are flags for options of nucleation scavenging, evaporative regeneration of aerosols, precipitation scavenging, dry deposition, and aerosol radiative impacts. There is also advective and diffusive transport. Further development of aerosol prediction in RAMS is in progress. ICLOUD is relevant only when LEVEL is set to 3.
IDRIZ integer	Similar to ICLOUD but controls the activation of drizzle rather than cloud. IDRIZ cannot = 5 unless ICLOUD =5. If IDRIZ > 0, then cloud droplets generally pass through the drizzle mode during growth before becoming rain. This slows autoconversion a bit. If IDRIZ = 5, then aerosols can nucleate directly to the drizzle mode if the median radius of the activated aerosol distribution > ~1.0 microns.
IRAIN integer	Flag that controls the model's treatment of rainwater and is relevant only if LEVEL is set to 3. If IRAIN is set to 0, rain is not activated, and any process involving the interaction of rain with other water species is not performed. Rain is activated when IRAIN is set to a value from 1 to 5. Activation means that the mixing ratio of rain is prognosed from conservation equations which include advective, diffusive, and precipitation tendencies, and source terms resulting from interactions between rain and other forms of water substance. The choice of values for IRAIN between 1 and 5 controls the manner in which the mean rain droplet diameter and number concentration are determined. With the flag equal to 1, the mean diameter is specified from a default value in the code, and the number concentration is diagnosed automatically from this mean diameter and the prognosed mixing ratio. When the flag is set to 2, the user specifies the mean diameter (in meters) in the namelist variable RPARM , and the number concentration is diagnosed as before. When the flag is set to 3, the user specifies the y-intercept value of the number concentration (this is the number concentration per unit diameter increment, (i.e., number per m ³ per m), evaluated at zero diameter, assuming a Marshall-Palmer (exponential) size distribution. The total number concentration (i.e., the total number of droplets per cubic meter) and the mean diameter are then both diagnosed from the y-intercept value and the prognosed mixing ratio. With a value of 4 for the flag, the user specifies the total number concentration of rain droplets (number per kg of air) directly in the namelist variable RPARM , and the mean diameter is automatically diagnosed from this concentration and the prognosed mixing ratio. When the flag is set to 5, the model activates a prognostic equation for rain droplet number concentration, and a special array for this quantity is added. Mean rain droplet diameter is then diagnosed from the prognosed mixing ratio and number concentration. Note that if any of the flags IRAIN , ISNOW , IAGGR , IGRAUP , or IHAILE is set to 5, the model will override settings of 1, 2, 3, or 4 for ALL of the others and make them 5. (Any that are set to zero will remain zero.) Thus, the model requires that if number concentration is predicted for any of these, it must be predicted for all of these species that are active. IPRIS is not in this list; it must ALWAYS be set to 5 (if pristine ice is activated). Independently of how flags for the other species are set. The choice of which value of IRAIN to use will depend to a large extent on how the user wishes to constrain the rain field, i.e., whether mean diameter, total number concentration, or both should be allowed to vary. For example, if the user is simulating an observed and documented precipitation event in which, say, mean droplet diameter was measured, it would make sense to specify that diameter as a fixed quantity. Independent selection of microphysical species to be activated versus not activated is controlled through the setting of IRAIN and the related parameters IPRIS , ISNOW , IAGGR , IGRAUP , and IHAILE described below. However, not all possible combinations are intended as valid, nor would they function properly without code modification. Activation of all categories simultaneously is the normal practice, although activation of the limited sets {cloud}, {cloud, rain}, {cloud, pristine ice}, {pristine ice}, {pristine ice, snow}, {pristine ice, aggregates}, and {pristine ice, snow, aggregates} are reasonable when only a limited range of microphysical processes need to be considered.
IPRIS integer	Similar to IRAIN but controls the activation of pristine ice rather than rain. The descriptions of the various settings given for IRAIN apply here, except that if pristine ice is activated at all its number concentration must be predicted as well. Hence, values of 1, 2, 3, and 4 are not permitted for IPRIS . If IPRIS = 5, RAMS assumes that ice forming nuclei (IFN) have a horizontally homogeneous, vertical profile of IFN that decreases with height. The IFN profile is a standard 3D scalar. Aerosol and IFN profiles can be modified in mic_init.f90. However, if IFN = 3, the default IFN profile is not used, but rather the model uses the population of aerosols with D > 0.5 microns as potential IFN . (See Saleeby and van den Heever 2013 and DeMott et al. 2010).
ISNOW integer	Similar to IRAIN but controls the activation of snow rather than rain. The descriptions of the various settings given for IRAIN apply here.
IAGGR integer	Similar to IRAIN but controls the activation of aggregates rather than rain. The descriptions of the various settings given for IRAIN apply here.

IGRAUP integer	Similar to IRAIN but controls the activation of graupel rather than rain. The descriptions of the various settings given for IRAIN apply here.
IHAIL integer	Similar to IRAIN but controls the activation of hail rather than rain. The descriptions of the various settings given for IRAIN apply here.
CPARM real	Similar to RPARM below but applies to cloud water rather than rain. If ICLOUD is set to 5 (see above), CPARM represents the concentration of cloud concentration nuclei (CCN) in number per kilogram of air. If ICLOUD is set to 7, a prognostic CCN field is activated in RAMS , and CPARM then represents the initial value (in number per kilogram of air) of the CCN concentration field.
DPARM real	Similar to RPARM , but applies to drizzle rather than rain.
RPARM real	Specifies either a fixed mean diameter (in meters), a fixed y-intercept number concentration per unit diameter increment (number per m ³ per meter), or a fixed total number concentration (in number per kg of air) to be imposed on the rain droplet field, in conjunction with settings 2, 3, and 4, respectively, for the flag IRAIN described above. If IRAIN is set to 0, 1, or 5, RPARM is ignored.
PPARM real	Obsolete since pristine ice number concentration must always be prognosed (if pristine ice is active).
SPARM real	Similar to RPARM , but applies to snow rather than rain.
APARM real	Similar to RPARM , but applies to aggregates rather than rain.
GPARM real	Similar to RPARM , but applies to graupel rather than rain.
HPARM real	Similar to RPARM , but applies to hail rather than rain.
GNU real array	Parameter with 7 values, one for each hydrometeor category in the order of cloud, rain, pristine ice, snow, aggregates, graupel, hail, drizzle. It specifies the shape parameter of the gamma distribution that all categories in the bulk microphysics submodel are assumed to follow. GNU = 1 indicates the Marshall-Palmer, or exponential, distribution, in which number concentration decreases monotonically with diameter throughout the size spectrum. Larger values of GNU indicate more general gamma distributions, in which the size distribution peaks at a positive diameter. The larger the value of GNU , the more narrowly distributed the spectrum is. The intent in using this variable is to specify a shape based on observation. The shape may in general depend on the type of precipitating system (deep convective, wintertime cyclone, etc.), as well as on the hydrometeor category (rain versus hail, for example). Little observational guidance is available to date on appropriate values of GNU . A Marshall Palmer distribution has been used in many models, although it may not be the closest of the gamma distribution family to real hydrometeor spectra. Sensitivity experiments with GNU in RAMS have demonstrated that it can sometimes have a significant effect on model results. The user is encouraged to experiment with the values of GNU , and to use observational guidance where possible. Ongoing research is aimed at determining appropriate values of GNU for each hydrometeor category in different weather situations. Values of 2 are suggested as middle-of-the-road values to start from, but it is sometimes argued that cloud spectra are generally narrower and should have gnu equal to at least 5.
HUCMFILE character	Is a single-valued character variable giving the directory path of the HUCM-SBM microphysics tables. The directory name is hardcoded as HUCM-SBM. By default, this directory is included in the "etc" directory of the model release. If you change the name or location of this directory, you need to make the appropriate change here in this flag. Used for LEVEL = 4 only.
NDTCOLL integer	Incremental number of timesteps to call collision-coalescence in the HUCM-SBM microphysics (LEVEL = 4 only). (Collision-coalescence is called every NDTCOLL timesteps).
IAEROSOL integer	Flag for number of specified CCN type aerosol categories 0, 1, 2, or 3. When turned on, arrays are allocated, and initial vertical profiles are set. Runs for LEVEL <= 3. [0] = no CCN aerosols. [1] = uses CCN1_MAX as near surface concentration. [2] = turns on a 2 nd mode and uses CCN2_MAX. [3] = turns on a 3 rd mode and uses CCN3_MAX. Vertical profiles can be customized in "src/micro/mic_init.f90".
ISALT integer	Flag for sea-salt aerosol categories. There are 3 modes: film drop, jet drop, and spume drop. Value of 0=off, 1=userset profiles, 2=sea-salt model. Runs for LEVEL <= 3.
IDUST integer	Flag for mineral dust aerosol categories. There are 2 modes: sub-micron and super-micron. Value of 0=off, 1=userset profiles, 2=dust lofting model. Can have soluble component for nucleation (AERO_EPSILON) but is insoluble (without soluble coating) for radiative purposes. Runs for LEVEL <= 3.

IABCARB integer	Flag for absorbing carbon aerosols categories. There are 2 modes: 1% black carbon and 2% black carbon. Both are treated similarly for nucleation, but differently for aerosol radiation. 2% BC is more absorbing. Value of 0=off, 1=userset profiles. Can have soluble component for nucleation (AERO_EPSILON) but is insoluble, without soluble coating, for radiative purposes. Runs for LEVEL <= 3.
IDUSTLOFT integer	Flag to choose which type of dust lofting to turn on. Only used if IDUST=2 . Value of 0 = Idealized lofting from any grid cell with erodible soil and low-vegetation, 1 = Paul Ginoux (2001) erodible fraction database and method. Runs regardless of LEVEL . All dust lofting schemes loft as a function of soil moisture and near surface wind speed. See (Saleeby and van den Heever (2013, JAMC) for details on idealized dust lofting scheme.
DUSTFILE character	Is a single-valued character variable giving the name (and path, if any) of the Dust lofting data file to use for surface dust erodible fraction. There are multiple file options in the default "etc" directory of the model release. The ones to choose begin with prefix "DustErodeFrac". If you change the name or location of these file, you need to make the appropriate change here in this flag.
ICCNLEV integer	Flag for aerosol microphysics treatment complexity. First note that when multiple aerosol categories are turned on, they will compete for nucleation based on the surface area of the fraction of particles that may nucleate. (see Saleeby and van den Heever 2013, JAMC) ICCNLEV=0 allows aerosol activation for nucleation but only nucleates additional aerosol in excess of the number of cloud droplets present in a given grid cell. Aerosols are not removed from the domain by nucleation. ICCNLEV=1 allows aerosol removal upon nucleation of new cloud droplets. Removal from each category starts at the large end of the distribution first. Largest aerosols are removed first, and the median radius is then recalculated before the next timestep and will be smaller since the larger aerosols were removed. ICCNLEV=2 is the same as ICCNLEV=1 but now we store the nucleated and removed aerosol mass in tracking arrays. A tracking array exists for each hydrometeor species. First, aerosol mass will enter the cloud category tracking array. As cloud is collected or undergoes any transfer to another category, a proportional amount of aerosol mass-within-cloud will be transferred as well. All possible transfers among hydrometeor species is accounted for. Upon hydrometeor evaporation, aerosols will be restored to a regenerated aerosol category. For regeneration, the mass of regenerated aerosols is proportional to the mass of evaporated hydrometeor mass. The number of restored aerosols equals the number of fully evaporated hydrometeors in each category. If the median radius of the regenerated aerosols is < 1 micron, then the aerosols enter the sub-micron regenerated category. If > 1 micron, they enter the super-micron regenerated category. Runs for LEVEL=3 only. NOTE: ICCNLEV=2 will function for single moment microphysics (except ICLOUD must = 5) but any regeneration will be suspect since number concentration may be diagnosed rather than predicted. This could lead to regeneration of unrealistic numbers of aerosols.
IIFN integer	Ice nucleation treatment level. IIFN=0 means no heterogeneous ice nucleation from ice nuclei. IIFN=1 uses the Meyers formula. IIFN=2 uses the DeMott et al. (2010,2013) formula. IIFN=1,2 use constant ice nuclei profiles set up using CIN_MAX as a maximum. IIFN=3 uses the DeMott formula based on the number concentration of aerosols > 0.5 microns in diameter. This is used in conjunction with ICCNLEV and responds accordingly depending on whether or not aerosols are removed upon nucleation. Note that if you use IIFN=3 and there are no aerosols > 0.5 micron in diameter, there will be not heterogeneous ice nucleation by condensation freezing or immersion freezing. Contact nucleation is still tied to Meyers scheme at this time. Haze nucleation is tied to the number concentration of aerosol categories 1, which is typically referred to as the general CCN category. Applies conditionally to LEVEL=3 or 4. IIFN=4 uses quasi-constant pristine ice + snow number concentration. Still needs IPRIS and ISNOW = 5. Quasi constant value is given by CIN_MAX in # per LITER . This differs from the other uses of CIN_MAX but is consistent with typical specs for ice crystal concentration in the literature.
IIFN_FORMULA integer	Flag to decide which of the DeMott et al. formulas to use for heterogeneous ice nucleation. If flag IIFN = 2 or 3, this flag is used. A value of 1 uses the DeMott et al. (2010) general formula, while a value of 2 uses the DeMott et al. (2014) dust aerosol formula.
IAERORAD integer	Aerosol-Radiation Model: 0 = Off, 1 = On. Runs regardless of LEVEL .
IAERODEP integer	Aerosol wet and dry deposition model: 0 = Off, 1 = On. Runs regardless of LEVEL . (precipitation scavenging or wet deposition of aerosols adds such mass to aerosol mass-in-hydrometeors tracking variables if iccnlev>=2)
IAEROPRNT integer	Print initial aerosol profiles: 0 = Off, 1 = On. Runs regardless of LEVEL .
IAEROHIST integer	Re-initialize aerosol profiles upon history restart: 0 = Off, 1 = On. Runs regardless of LEVEL .
CIN_MAX real	Maximum (lowest model level) ice nuclei concentration (#/mg) used to generate initial vertical profile. Only used if IIFN=1or2 . See subroutine "init_ifn" in file "mic_init.f90" for IFN/INP profile initialization. A couple of special INP profile specifications are available by looking in the initialization code noted above. For, IIFN=4 the CIN_MAX value is a special case for quasi-constant ice concentration with special units of (#/L).

CCN1_MAX real	Maximum (lowest model level) aerosol mode-1 concentration in #/mg used to generate initial vertical profile. Only used if IAEROSOL >= 1.
CCN2_MAX real	Maximum (lowest model level) aerosol mode-2 concentration in #/mg used to generate initial vertical profile. Only used if IAEROSOL >= 2.
CCN3_MAX real	Maximum (lowest model level) aerosol mode-3 concentration in #/mg used to generate initial vertical profile. Only used if IAEROSOL == 3.
DUST1_MAX real	Maximum (lowest model level) small dust mode concentration used to generate initial vertical profile. Only used if IDUST =1.
DUST2_MAX real	Maximum (lowest model level) large dust mode concentration used to generate initial vertical profile. Only used if IDUST =1.
SALTF_MAX real	Maximum (lowest model level) film drop salt mode concentration used to generate initial vertical profile. Only used if ISALT =1.
SALTJ_MAX real	Maximum (lowest model level) jet drop salt mode concentration used to generate initial vertical profile. Only used if ISALT =1.
SALTF_MAX real	Maximum (lowest model level) spume drop salt mode concentration used to generate initial vertical profile. Only used if ISALT =1.
ABC1_MAX real	Maximum (lowest model level) absorbing carbon 1st mode concentration used to generate initial vertical profile. Only used if IABCARB =1. Mode 1 is 1% BC and 99% OC.
ABC2_MAX real	Maximum (lowest model level) absorbing carbon 2nd mode concentration used to generate initial vertical profile. Only used if IABCARB =1. Mode 2 is 2% BC and 98% OC.
IAERO_CHEM integer array	Aerosol chemistry type (1 = ammonium sulfate, 2 = sodium chloride). Input value for each aerosol category. The categories in order are: 1. Mode-1 general aerosol, 2. Mode-2 general aerosol, 3. Mode-3 general aerosol, 4. Small dust, 5. Large dust, 6. Film mode sea salt, 7. Jet mode sea salt, 8. Spume mode sea salt, 9. Absorbing carbon (1% Black Carbon), 10. Absorbing carbon (2% Black Carbon), 11. Sub-micron regenerated aerosol, 12. Super-micron regenerated aerosol.
AERO_EPSILON real array	Fraction of soluble material on aerosols. Value is 0.0 -> 1.0. Input value for each aerosol category.
AERO_MEDRAD real array	Aerosol log-normal distribution median radius. Values in the model are forced between 0.01 microns and 6.5 microns. Input value for each aerosol category.
ITRKEPSILON integer	Aerosol tracking flags: 0=off, 1=on. Tracking aerosol solubility for regeneration. (overrides default solubility of regenerated aerosol categories in AERO_EPSILON)
ITRKDUST integer	Aerosol tracking flags: 0=off, 1=on (diagnostic only). Tracking dust sub-species mass mixing ratio within hydrometeors.
ITRKDUSTIFN integer	Aerosol tracking flags: 0=off, 1=on (diagnostic only) Tracking mass mixing ratio of dust that served as ice nuclei.

\$MODEL_SOUND Namelist

The **\$MODEL_SOUND** namelist consists of a set of variables for specifying a sounding to be used in initializing a simulation. This method of initialization is performed horizontally homogeneously and is performed only when the flag **INITIAL** is set to 1. The variables in the **\$MODEL_SOUND** namelist consist mainly of (1) arrays containing the actual values of pressure or height, velocity, temperature, and moisture, and (2) a set of flags specifying how the array values are to be interpreted. An alternative form of specifying five of the variables in this namelist, **PS**, **TS**, **RTS**, **US**, and **VS**, all described below, is to enter them into a file called **SOUND_IN** in tabular form. This option is selected by setting the first value of **PS** to 0. In this case, the model opens and reads the file **SOUND_IN**, which must reside in the same directory as the model executable. The code that reads from this file is contained in the file *rhhi.f90* in subroutine **ARRSND** in the first DO loop. Free format is assumed with one sounding level per record starting with the lowest sounding level at the top of the file, and the five variables are read from each record in the order shown above.

Variable name	Description
IPSFLG integer	Is a flag specifying how the values given for PS are to be interpreted. When IPSFLG is set to 0, all values in PS are interpreted as pressures in millibars, and when IPSFLG is set to 1, all values in PS are interpreted as heights in meters. In either case, the PS values are used to define the heights in the sounding at which all sounding data

	applies.
ITSFLG integer	Is a flag specifying how the values given for TS are to be interpreted. When ITSFLG is set to 0, all values in TS are interpreted as temperatures in degrees C, when ITSFLG is set to 1, all values in TS are interpreted as temperatures in degrees K, and when ITSFLG is set to 2, all values in TS are interpreted as potential temperatures in degrees K. The options allow the model to read sounding temperature data in any of these forms.
IRTSFLG integer	Flag specifying how the values given for RTS are to be interpreted. When IRTSFLG is set to 0, all values in RTS are interpreted as dew point temperatures in degrees C, when IRTSFLG is set to 1, all values in RTS are interpreted as dew point temperatures in degrees K, when IRTSFLG is set to 2, all values in RTS are interpreted as water mixing ratio values in grams per kilogram, when IRTSFLG is set to 3, all values in RTS are interpreted as relative humidity in percentage, and if IRTSFLG is set to 4, all values in RTS are interpreted as dew point depressions in degrees K. The options allow the model to read sounding moisture data in any of these forms.
IUSFLG integer	Flag specifying how the values given for US and VS or for USNDG and VSNDG are to be interpreted. When IUSFLG is set to 0, all values in US (or USNDG) are interpreted as components of velocity in meters per second in the x-direction, and all values in VS (or VSNDG) are interpreted as components of velocity in meters per second in the y-direction. When IUSFLG is set to 1, all values in US (or USNDG) are interpreted as wind directions in degrees azimuth clockwise from true north, and all values in VS (or VSNDG) are interpreted as wind speeds in meters per second.
IO3FLG integer	If = 0, ozone profile given from McLatchy. If = 1, ozone profile included in RAMSIN namelist or SOUND_IN in ppb.
SOUND_FILE char array	Input sounding files name with full path. Only used if PS(1)=0.
HS real array	Sounding heights, but only the first value is read from the namelist. This first value indicates the absolute height (relative to sea level) of the first or lowest sounding level. It must be at or below the lowest topography height in the RAMS model domain. This may require adding one or more levels to the bottom of a real sounding.
PS real array	Pressures in millibars or the heights in meters of the sounding levels starting at the ground and proceeding upward. The choice of meaning is controlled by the value of IPSFLG . If PS represents heights, the first value of PS is still specified as a pressure in millibars, and its corresponding height is read from namelist variable HS . The primary requirement of the sounding data is that it extend to a height greater than the top of the model domain, and at least as low as the lowest topography in the domain. Thus, the top sounding level must contain either a height or a pressure higher in the atmosphere than the model domain top, which is itself determined by the combination of namelist parameters NNZP , DELTAZ , DZRAT , and DZMAX , or by ZZ . A maximum of 200 values may be specified for PS , although with code modification this number could be increased if necessary. The values given for PS will determine the vertical spatial resolution of the sounding and should be chosen to properly represent any important significant levels. Note the special option of setting the first value of PS to 0., as described at the beginning of this Section.
TS real array	Sounding temperature values. The way in which the values are interpreted is specified by namelist variable ITSFLG . The number of values specified for TS must equal the number of values specified for PS . The user should use caution in specifying sounding temperatures. Superadiabatic profiles in the sounding are strongly discouraged (which does not mean that the model simulation may not successfully develop superadiabatic regions after initialization). High vertical resolution is sometimes desirable in a sounding because the model fields of temperature and moisture mixing ratio are interpolated linearly from it in the vertical direction, and relative humidity is a very nonlinear function of these two interpolated fields. This can result in supersaturated layers in the model initial fields where none exist in the sounding.
RTS real array	Sounding moisture values. The way in which the values are interpreted is specified by namelist variable IRTSFLG . The number of values specified for RTS must equal the number of values specified for PS . (Also see TS .)
US real array	Sounding wind values. The way in which the values are interpreted is specified by namelist variable IUSFLG . The number of values specified for US must equal the number of values specified for PS .
VS real array	Sounding winds values. The way in which the values are interpreted is specified by namelist variable IUSFLG . The number of values specified for VS must equal the number of values specified for PS .
O3S real array	Sounding ozone profile in ppb. Same length as other input sounding profiles.

RAMS ISAN Configuration Parameters

The following configuration parameters are defined in the *isan_coms.f90* file. They function as dimensions for several arrays and are responsible for the maximum values that can be set for some namelist variables during execution. These are mentioned for completeness and would not normally need to be changed except in extreme circumstances. They may be set to fairly large values in a standard model installation, as their settings do not affect total memory usage very much; most of the memory is allocated dynamically.

MAXPR (int)	Maximum number of vertical levels that can be used in the pressure data.
MAXISN (int)	Maximum number of vertical levels that can be used in the isentropic analysis.
MAXX (int)	Maximum number of grid points (in the "x" or east-west direction) in the pressure grids.
MAXY (int)	Maximum number of grid points (in the "y" or north-south direction) in the pressure grids.
MAXTIMES (int)	Maximum number of data analysis times that can be processed in a single run.
MAXAGRDS (int)	Maximum number of RAMS grids that can have varfiles generated.
MAXSIGZ (int)	Maximum number of vertical levels that can be used in the σ_z analysis.
MAXLEV (int)	Maximum number of levels in an input rawinsonde.
MAXSNAME (int)	Maximum number of input observations
MAXISFILES (int)	Maximum number of input data times

RAMS ISAN Namelists

Variable name	Description
ISZSTAGE integer	Switch with a value of 0 (no) or 1 (yes), indicating whether the isentropic/ σ_z stage of ISAN is to be run. This stage inputs observational upper air datasets, as prepared separately in 'RALPH2 format' and objectively analyzes the data onto both isentropic and σ_z surfaces. It also inputs surface observations and objectively analyzes them onto a surface that is itself an objectively analyzed height surface based on surface elevations of the surface stations. This stage and the subsequent 'varfile' stage (see IVRSTAGE) can be run at once or each can be executed individually. The isentropic stage outputs the isentropic and σ_z files which the varfile stage will input if run separately.
IVRSTAGE integer	Similar to ISZSTAGE (see above) but applies to the 'varfile' stage, which produces varfiles for model initialization from isentropic and σ_z fields generated by the isentropic/ σ_z stage of ISAN.
ISAN_INC integer	Desired interval between consecutive data processing times. Certain files may be available at more frequent times than others, as, for example, surface observations in contrast with rawinsonde observations. Setting ISAN_INC to the standard 12-hour interval between rawinsondes will cause surface observations to be processed at only the rawinsonde times. The beginning and duration of the time period to process observational data are specified in IYEAR1 , IMONTH1 , IDATE1 , ITIME1 , and TIMMAX in the \$GRIDS namelist. The format for ISAN_INC is <i>hhmm</i> .
I1ST_FLG integer	What choice the model should make if the first guess field is not found but was requested to be used. This situation may arise, for example, in an operational setting where observational data are expected periodically from an outside source, but where something has prevented the data from being available. If I1ST_FLG is set to 1, it instructs ISAN to skip all processing of this data time and proceed with the next time. This will result in a particular varfile not being generated. If I1ST_FLG is set to 2, it instructs ISAN to stop.
IUPA_FLG integer	Similar to I1ST_FLG , but applies to input rawinsonde data instead of the first guess field. Another difference: if IUPA_FLG is set to 3, ISAN is instructed to process the given data time without rawinsonde data.
ISFC_FLG integer	Similar to I1ST_FLG , but applies to input surface data instead of the first guess field. Another difference: If ISFC_FLG is set to 3, ISAN is instructed to process the given data time without surface data.
IDATAIN integer	Flag for gridded dataset that you used to make the DPREP files (see IAPR). IDATAIN =0 indicates that relative humidity is the moisture variable in the DPREP files, while IDATAIN =1 indicates that specific humidity is used instead. A conversion is told to be done at the ISAN stage for IDATAIN =1 in order to obtain relative humidity for RAMS. (Be aware of what data is contained in the DPREP files named in IAPR below. For example, GDAS-FNL typically contains RH, NARR contains specific humidity, and ERA5 has both available. (It is suggested to use specific humidity with ERA5 data due to complex specification of RH in ERA5 that is not compatible with RAMS.)
IAPR character	Filename prefix, with directory path if applicable, of the input pressure files that were generated by dataprep. All files that have this prefix (and that are within the specified processing parameters) will be processed.
IARAWI character	Filename prefix, with directory path if applicable, of the input rawinsonde files that were generated by dataprep. All files that have this prefix (and that are within the specified processing parameters) will be processed.
IASRFCE character	Filename prefix, with directory path if applicable, of the input surface observation files that were generated by dataprep. All files that have this prefix (and that are within the specified processing parameters) will be processed.

VAR_HFILE character	Analysis <i>state</i> files to be read for use in the history file MAKEHFILE scheme. The character string placed in this variable must be the full name of the <i>state</i> header file, including the appropriate path. However, only the prefix is used to determine the set of <i>state</i> files to be used. Further details on the standard syntax of the analysis file names are given in the description of AFILOUT . This option can be expanded to include whatever history file variable you may wish to nudge.
VARPFX character	Filename prefix, with directory path if applicable, of two classes of files that are output from ISAN. The first file type is the 'varfile', which is the final stage of processing by ISAN and contains velocity, pressure, potential temperature, and vapor mixing ratio fields interpolated to the model grid(s) and ready for initialization and/or time-dependent data assimilation. The other file type is the 'isentropic/ σ_z file', which contains the output from the isentropic, σ_z , and surface analyses that are performed before the varfile stage of ISAN (see ISZSTAGE). While both file types are assigned names that begin with the prefix plus the year, month, date, and time automatically appended, the varfile name is additionally given a 'V' character, while the isentropic/ σ_z file name is given an 'I' character. One run of ISAN may process data at one or many analysis times, and files for all times are named with the same prefix but different time designations.
IOFLGISZ Integer	Flag indicating whether to write isentropic/ σ_z files: 0 – no, 1 – yes. (See ISZSTAGE and VARPFX).
IOFLGVAR integer	Flag indicating whether to write out the varfiles: 0 – no, 1 – yes. (See ISZSTAGE and VARPFX).

\$ISAN_ISENTROPIC Namelist - ISAN Isentropic/ σ_z Stage

Variable name	Description
NISN integer	Number of isentropic levels on which to perform objective analysis. A value around 40 is suggested for NISN to provide adequate vertical resolution.
LEVTH integer array	Isentropic levels in integer degrees Kelvin of the isentropic grid on which objective analysis is performed. The spacing that normally provides sufficient vertical resolution is 1-2 K near the ground, 3-5 K in the remainder of the troposphere, 20-30 K in the lower stratosphere, and 50 K in the middle stratosphere. It is now common practice in RAMS to use the NCEP reanalysis pressure level data as the first guess field in RAMS , and the top level of this dataset is 10 mb. Thus, isentropic levels may be specified to values as high as 800 K, in case the model domain is unusually high.
NIGRIDS integer	Number of RAMS grids to analyze. This must be less than or equal to NIGRIDS in the \$MODEL_GRIDS namelist. Although the ability exists to analyze atmospheric data and generate a varfile independently for each RAMS grid, this sometimes does not produce analyses that match smoothly across nested grid boundaries. Thus, care should be taken when using varfiles for nested grids for 4DDA (Newtonian nudging during model runtime) (see TNUDCENT). Gradients of prognostic variables initialized at grid boundaries will usually weaken with time if they are not continuously restored by strong nudging.
TOPSIGZ real	σ_z analysis will be done to all model levels under this height (in meters).
HYBBOT real	The hybrid analysis between the isentropic and σ_z datasets is accomplished in this version of ISAN by a weighted blending of the data over a specified, terrain-following layer using a linear (in height above the surface) weighting profile. At the bottom of the layer, the data is completely from the σ_z analysis; at the top of the layer, the data is completely from the isentropic analysis. HYBBOT is the approximate height above the surface of the bottom of this layer. The actual bottom is found as the closest model level to HYBBOT .
HYBTOP real	Approximate height above the surface of the top of the isentropic/ σ_z blending layer (see HYBBOT above). The actual height of the top is found as the closest model level to HYBTOP .
SFCINF real	In addition to the blending of the isentropic and σ_z analysis in the creation of the varfile, the surface data analysis is also blended in a layer near the ground. This only occurs if there are actual surface observations close enough to a grid point and the model level is within a certain vertical distance from the actual station height. This distance is SFCINF (meters). The actual weight given to the surface analysis varies from a full weight if the difference between the model level and station height is 0. to no weight for the surface (full weight for the upper air data) when the difference reaches SFCINF .
SIGZWT real	Optional weight given to the σ_z analysis in the blending process. If SIGZWT is set to 0., then the varfile will only contain information from the isentropic analysis. If SIGZWT is set to 1., then the varfile will contain full information from the σ_z analysis under HYBBOT . Any value between 0. and 1. may be used. For most applications, this should be set to 1.
NFEEDVAR integer	In the creation of the varfiles when more than one grid is analyzed, the data can optionally go through the nesting "feedback" process. The feedback must be done if the 4-dimensional data assimilation options are to be used. Set NFEEDVAR to 1 to do the feedback. Set NFEEDVAR to 0 usually only if you are doing a data analysis on different grid resolutions, will not be running the model, and you want to see the unmodified results

	of the different resolution analyses. Normally, leave this set to 1.
MAXSTA integer	Maximum number of rawinsondes expected in the data access. Memory will be allocated using this number, so don't set it too big.
MAXSFC integer	Maximum number of surface observations expected in the data access. Memory will be allocated using this number, so don't set it too big.
NOTSTA integer	Number of stations in the domain area to be excluded if they are found. The specific stations to be excluded are indicated by NOTID (see below). Excluding stations in this way is the means for eliminating stations with bad data.
NOTID char array	Station IDs to be excluded from any further processing (see NOTSTA above). Prefix the station ID with an 'r' for rawinsonde or an 's' for surface observation.
USED_FILE character	Filename prefix used to output stations actually used in the analysis.
IOBSWIN integer	To define the variables: obssecs - absolute time of an observation (in seconds) nasecs - absolute time of the data analysis (in seconds) IOBSWIN is the namelist parameter to set. There are three conditions: - If IOBSWIN equals zero, then obssecs must exactly equal nasecs. Otherwise, the observation will not be used. - If IOBSWIN is less than zero, then the time of the observation must be in the window from (nasecs-abs(iobswin)) to nasecs. - If IOBSWIN is greater than zero, then the time of the observation must be in the window from (nasecs-iobswin) to (nasecs+iobswin). So, for example, if your data analysis time is 1200 UTC: - if you set IOBSWIN=-3600, the observation time must be between 1100-1200 UTC to be used - if you set IOBSWIN=3600, the observation time must be between 1100-1300 UTC to be used
STASEP real	Surface observations will be objectively analyzed with the Barnes scheme that assumes uniform data coverage. Frequently, there are 2 surface stations reporting in the same city. As the surface stations are processed, any station within a distance STASEP (in degrees of latitude) of a previously processed station will be discarded unless the new one has less missing data than the previous one, in which case the other previous station will be discarded.
IGRIDFL integer	Flag which controls blending of gridded pressure level data and observations. If IGRIDFL = 0, no pressure grid point data is used, only observations. If IGRIDFL = 1, all observations (those not discarded as specified by the STASEP and NOTSTA parameters - see above) and all pressure grid point data are used. If IGRIDFL = 2, all observations (those not discarded) are used, but pressure data are used only from geographic locations that (1) have no sounding closer than the distance specified in GOBRAD and (2) do not have soundings in 3 or 4 quadrants (defined by intersecting east-west and north-south lines) all closer than the distance specified by GOBSEP (see GOBRAD and GOBSET). If IGRIDFL = 3, only gridded pressure level data are used, no observations. If IGRIDFL = 4, the gridded pressure data is taken as a first guess field. The observations are objectively analyzed and applied to this first guess field as deviations.
GRIDWT integer array	The implementation of the Barnes scheme in RAMS allows differential weighting of any station or pressure data point to give it more or less influence compared to other stations. GRIDWT has been implemented to reduce the weight of the gridded pressure data set relative to the actual observations. For example, if a rawinsonde observation were made at the same location as a gridded pressure point, a setting of GRIDWT = .1, would give the rawinsonde 10 times more weight than the pressure data. However, remember that the gridded data now are interpolated to the actual RAMS grids, so that there are many more pressure grid points than soundings. Also, there is a pressure grid point at the exact location of the grid to be analyzed. Therefore, a much lower setting of GRIDWT will usually be optimal. The actual value will take some experimentation but a value in the .1 to .001 may be appropriate. Note also that the value is grid dependent and must have as many values specified as NIGRIDS. In general, GRIDWT will become larger as the grid spacing of the RAMS grid increases.
GOBSEP real	If IGRIDFL (described above) is set to 2, a pressure data grid point will not be used in the objective analysis if it is within GOBSEP degrees of an upper air observation.
GOBRAD real	If IGRIDFL (described above) is set to 2, a pressure data grid point will not be used in the objective analysis if there are observations in three quadrants of a circle of GOBRAD radius (degrees). The circle is divided into quadrants by north-south and east-west diameters.
WVLNTH	In the Barnes objective analysis scheme, a degree of smoothing to be applied is determined from two

real array	parameters. The first, WVLNTH , is the wavelength of the data on the isentropic and σ_z surfaces (upper air) specified in kilometers to be retained. The second, RESPON (see below) is the fractional amplitude at which to retain that wavelength. Responses of other wavelengths are determined given a value of .3 set for the "gamma" parameter in the Barnes scheme. Note that WVLNTH is grid dependent and there must be as many values specified as NIGRIDS .
SWVLNTH real array	Similar to WVLNTH (described above) but pertains to surface rather than upper air data and analysis. It uses the same RESPON value as WVLNTH . Note that SWVLNTH is grid dependent and there must be as many values specified as NIGRIDS . For the surface objective analysis. Note that it is grid dependent and there must be as many values specified as NIGRIDS .
RESPON real array	Percentage of specified wavelength amplitudes to be retained in both the upper air and surface analyses (see WVLNTH and SWVLNTH above). Note that RESPON is grid dependent and there must be as many values specified as NIGRIDS .

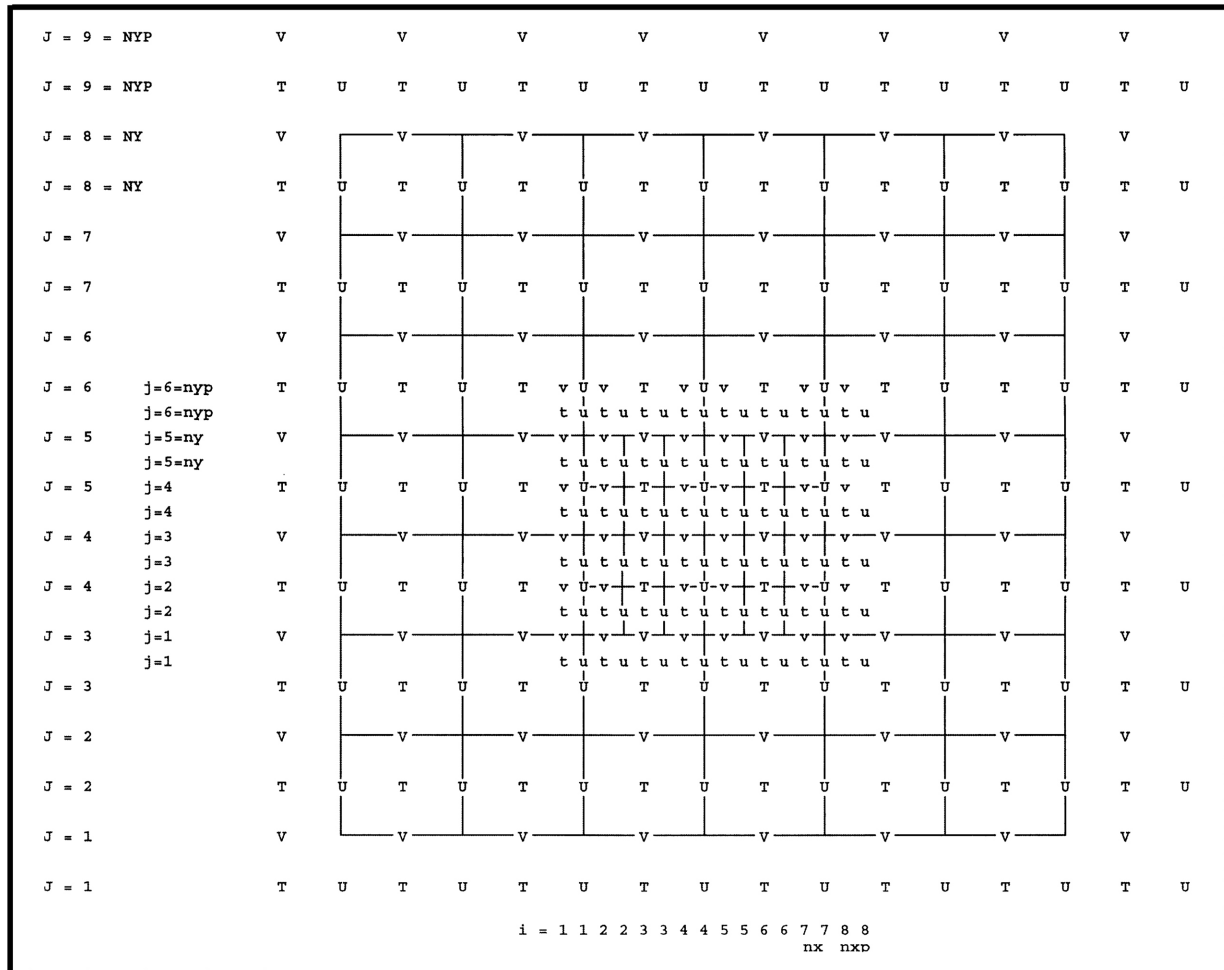
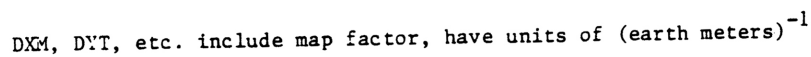


Figure 1: Horizontal Arakawa-C grid stagger.


$$\text{sigma-z} \rightarrow z^* = H \left[\frac{z - z_0}{H - z_0} \right] = \left[\frac{z - z_0}{1 - \frac{z_0}{H}} \right]$$

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